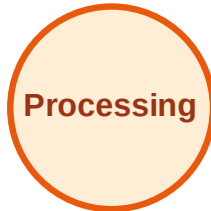


BFS Traversal

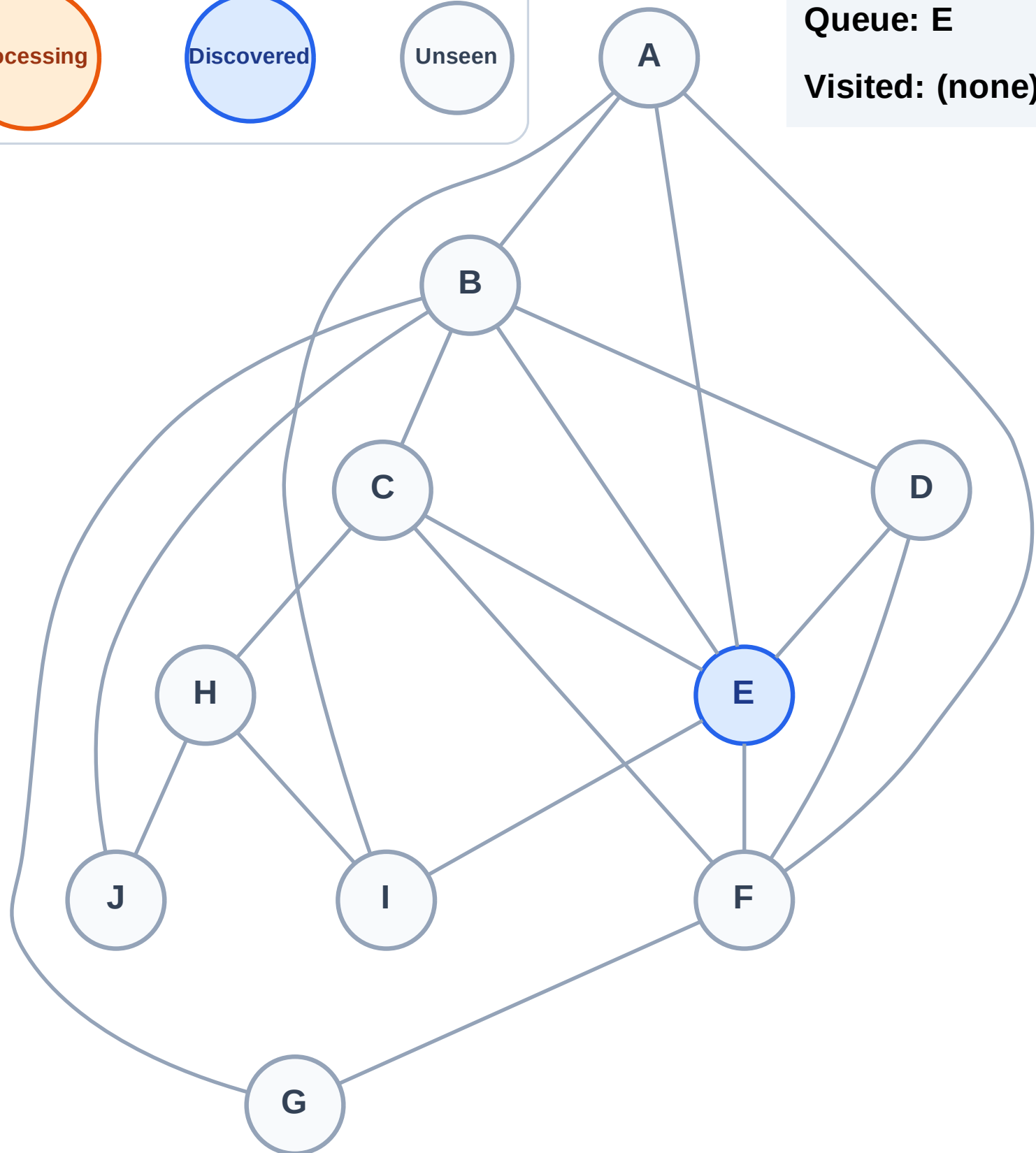
Step 1: Start at E

Legend



Queue: E

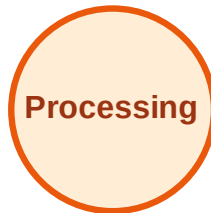
Visited: (none)



BFS Traversal

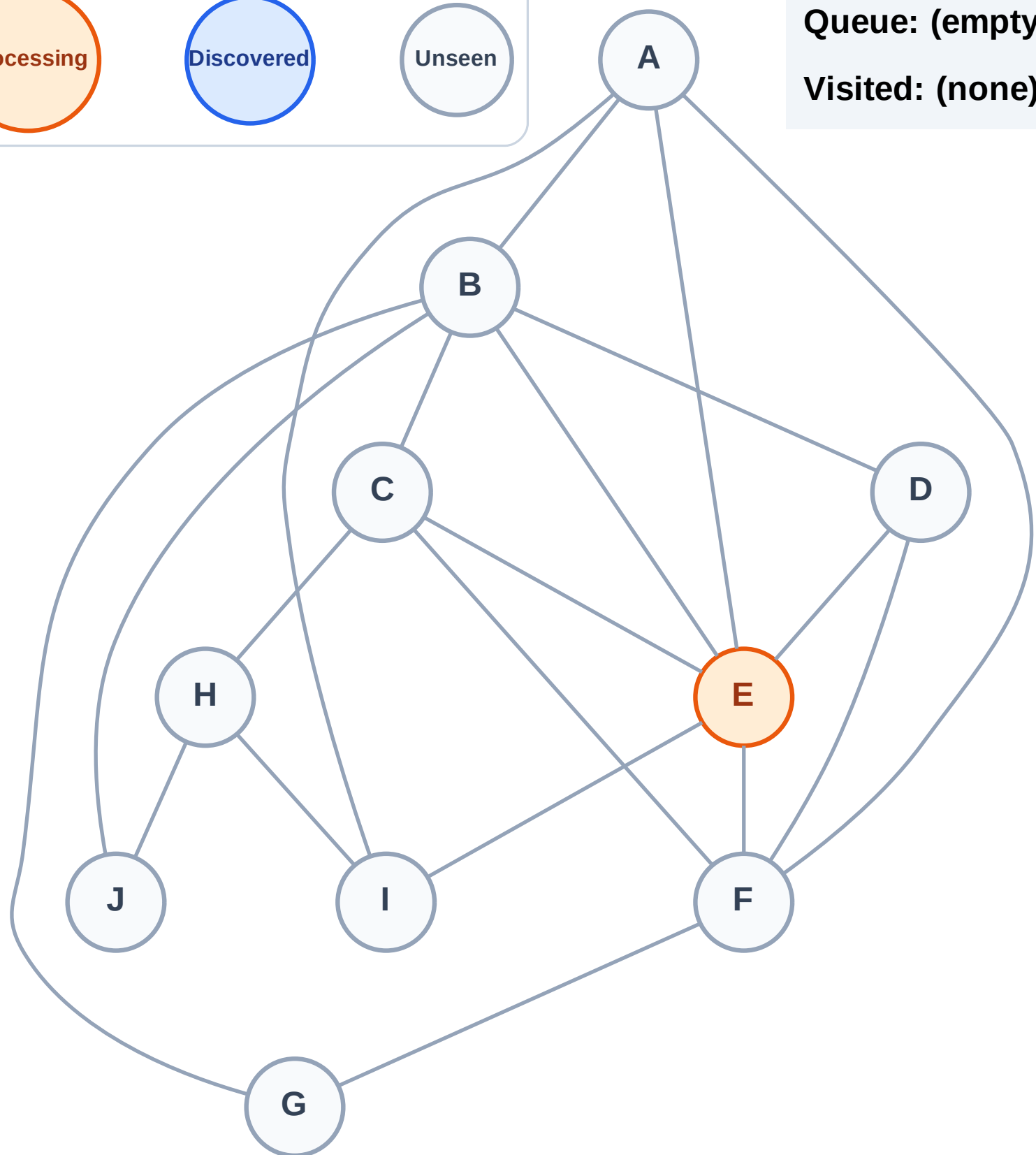
Step 2: Process node E

Legend



Queue: (empty)

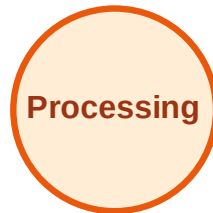
Visited: (none)



BFS Traversal

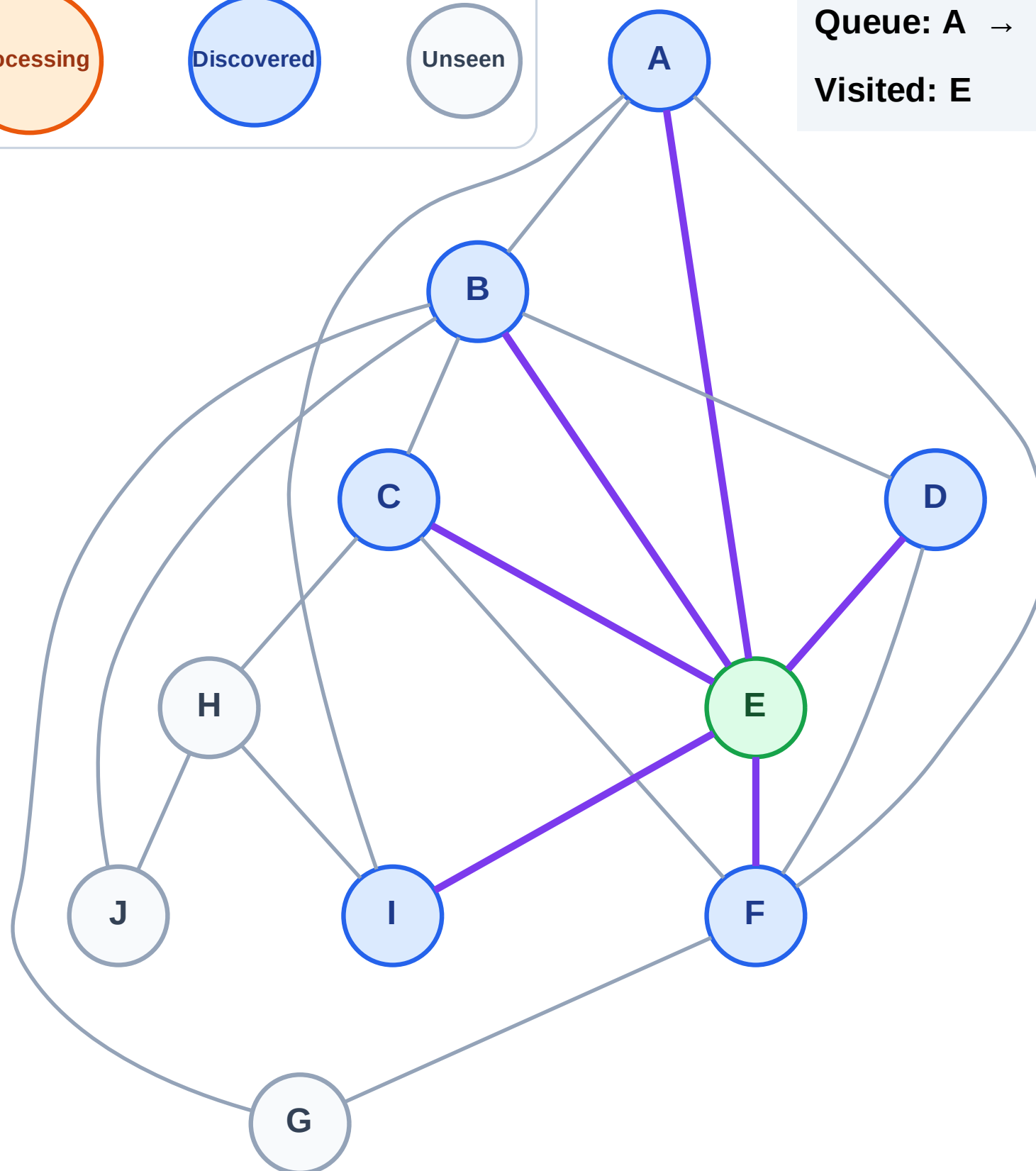
Step 3: Discover A, B, C, D, F, I from E

Legend



Queue: A → B → C → D → F → I

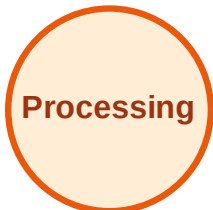
Visited: E



BFS Traversal

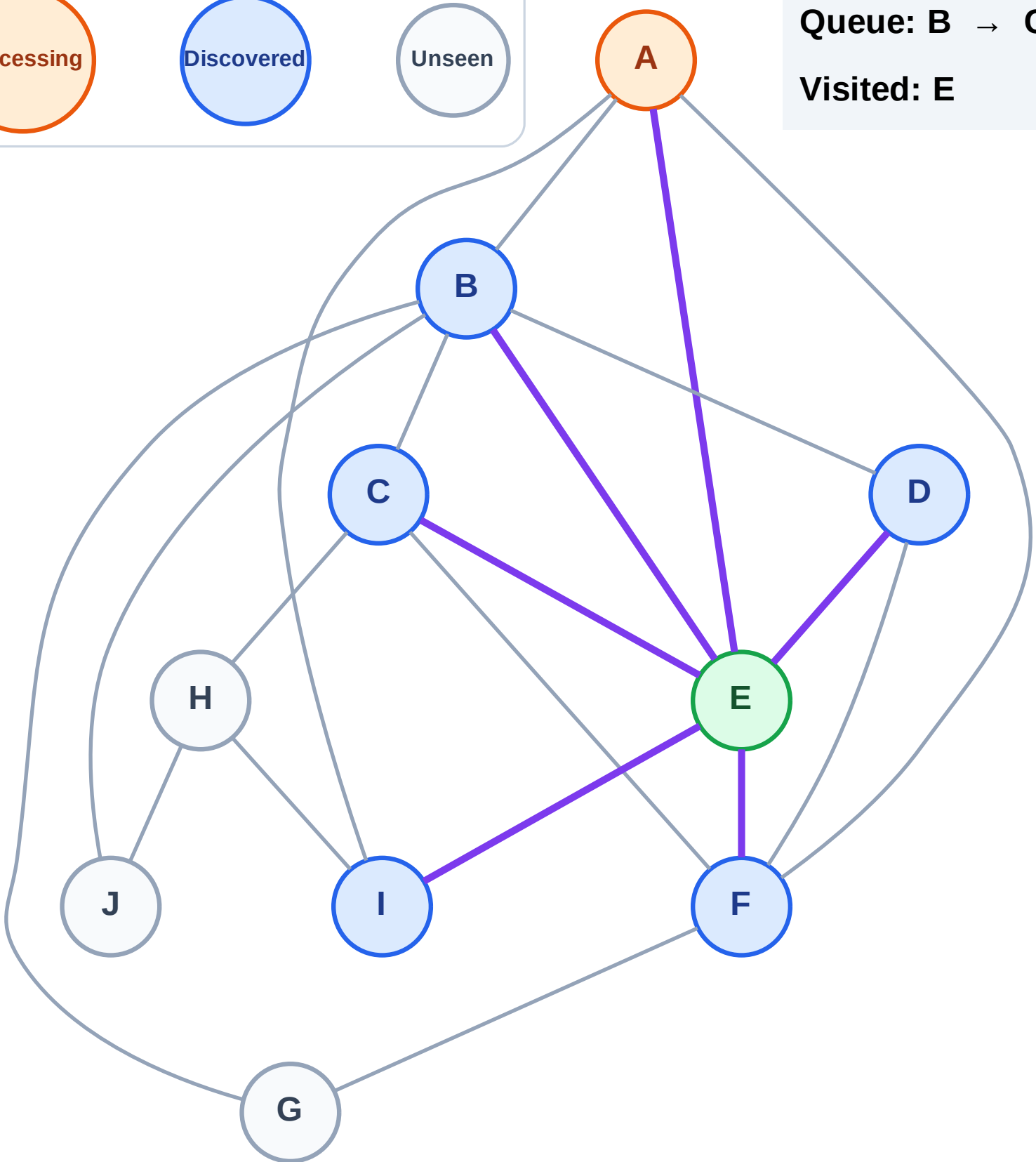
Step 4: Process node A

Legend



Queue: B → C → D → F → I

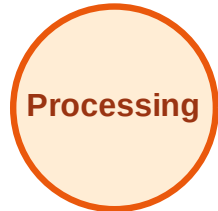
Visited: E



BFS Traversal

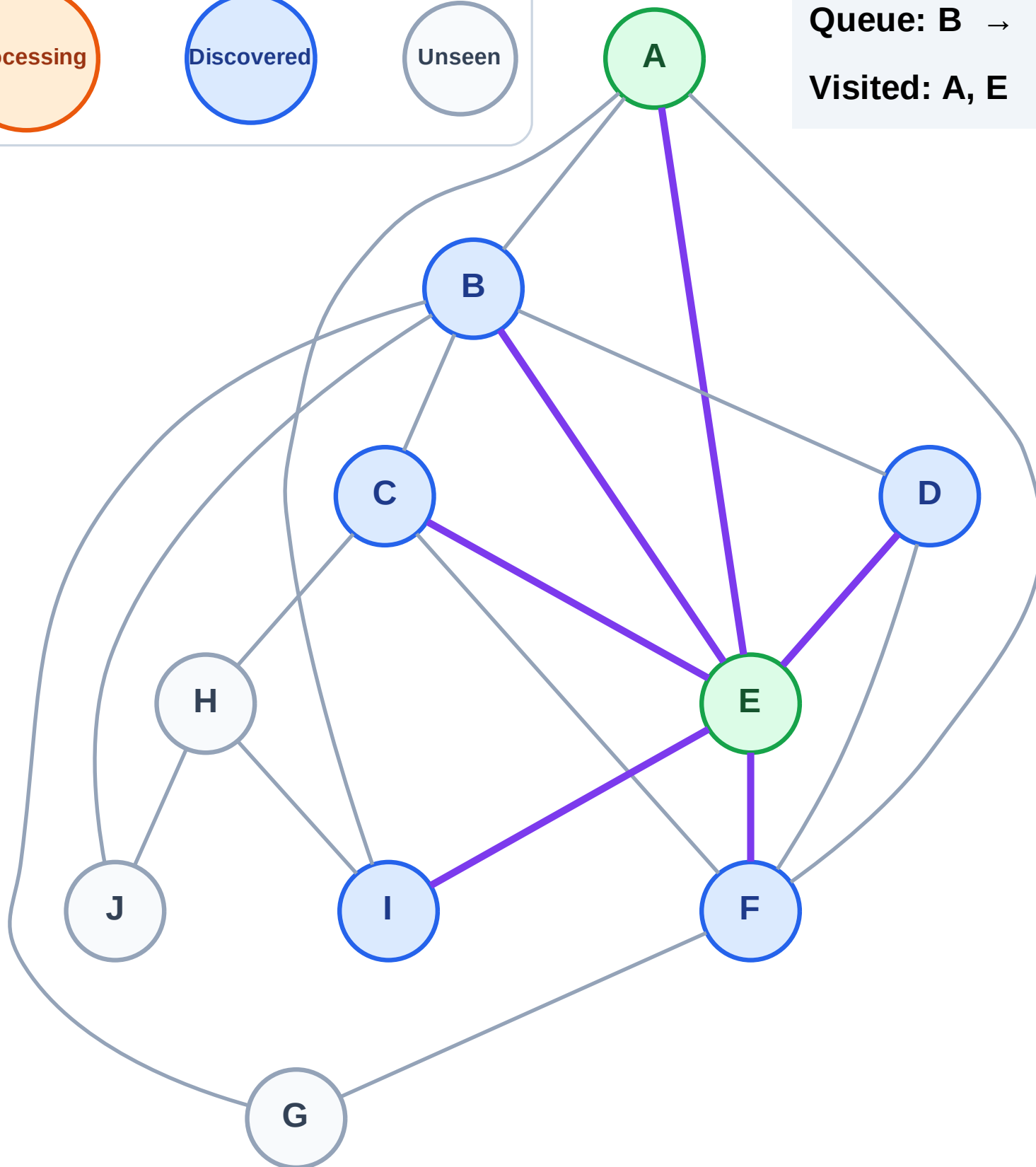
Step 5: No new neighbors from A

Legend



Queue: B → C → D → F → I

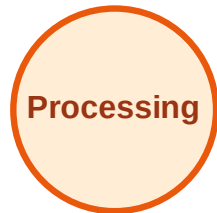
Visited: A, E



BFS Traversal

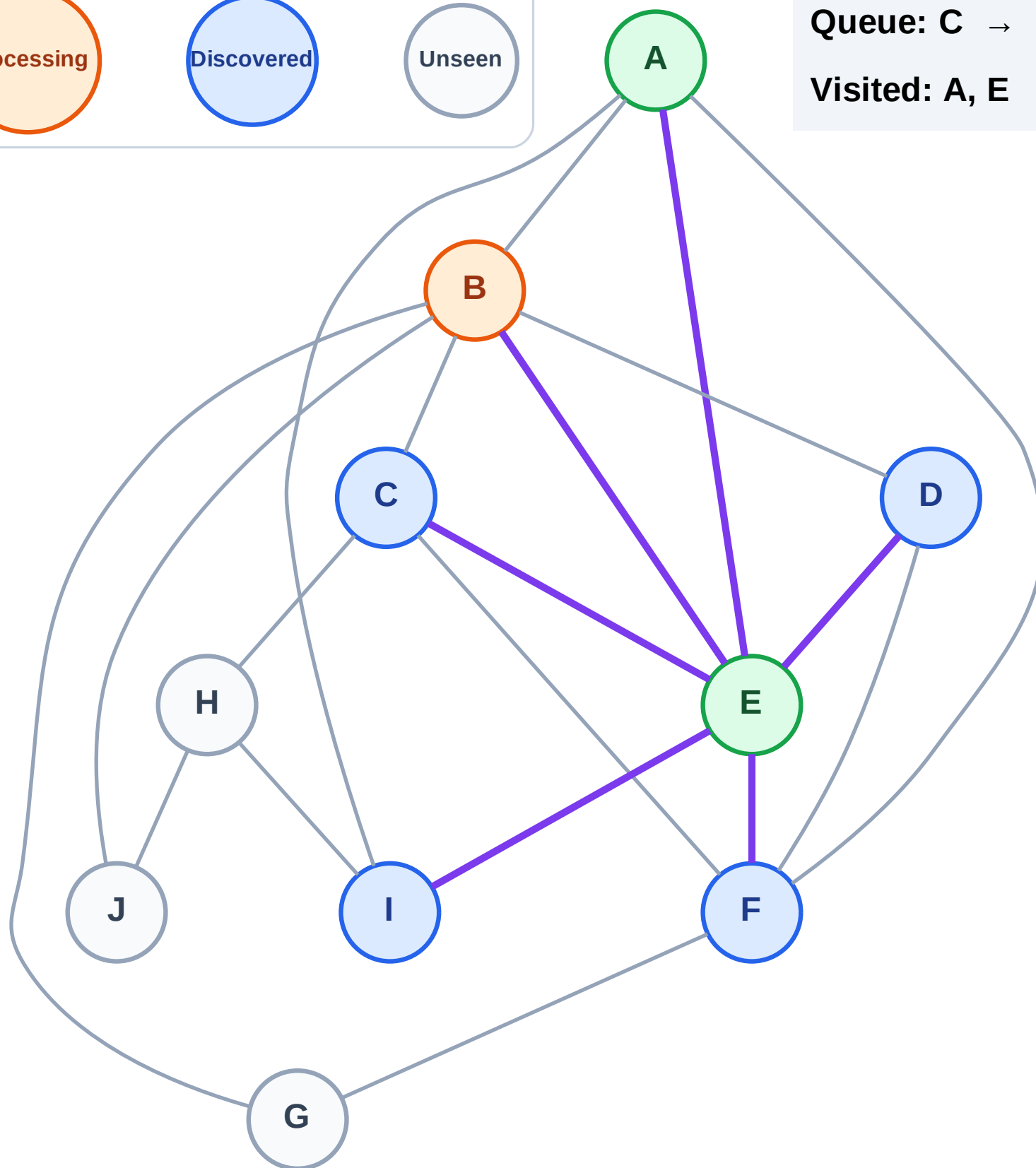
Step 6: Process node B

Legend



Queue: C → D → F → I

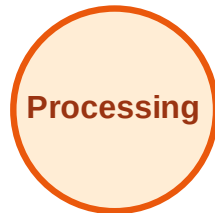
Visited: A, E



BFS Traversal

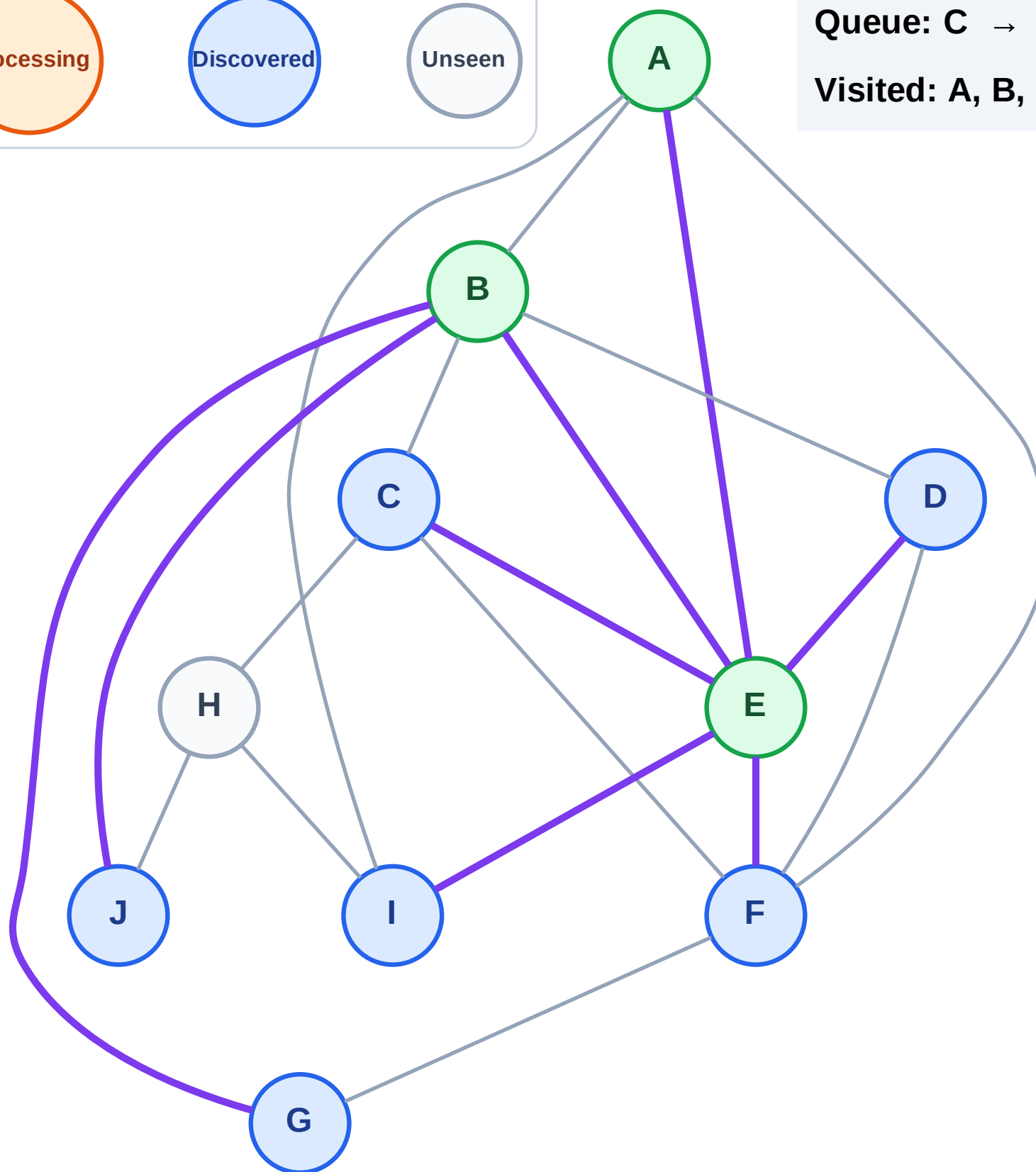
Step 7: Discover G, J from B

Legend



Queue: C → D → F → I → G → J

Visited: A, B, E



BFS Traversal

Step 8: Process node C

Legend

Complete

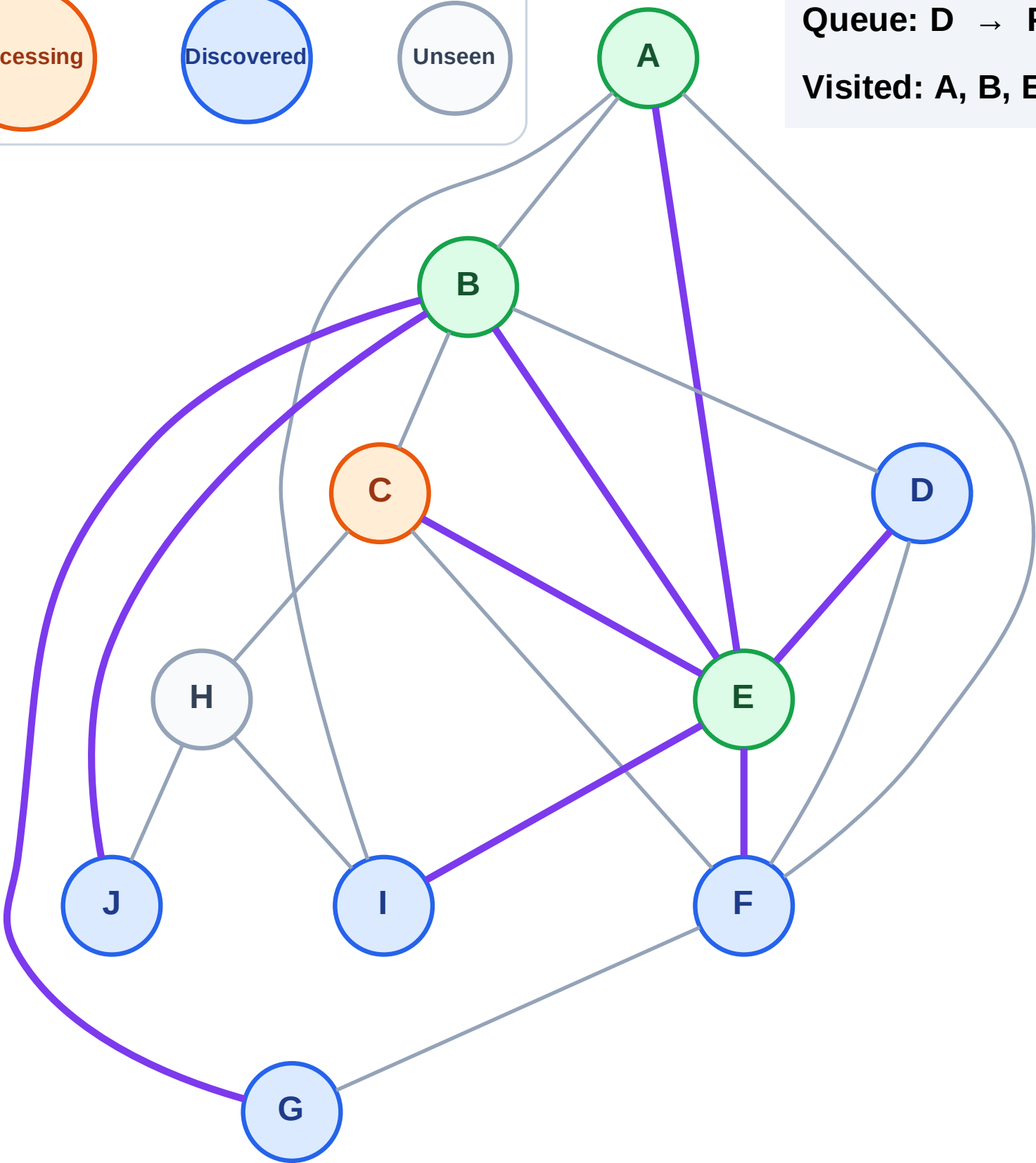
Processing

Discovered

Unseen

Queue: D → F → I → G → J

Visited: A, B, E



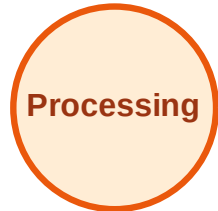
BFS Traversal

Step 9: Discover H from C

Legend



Complete



Processing



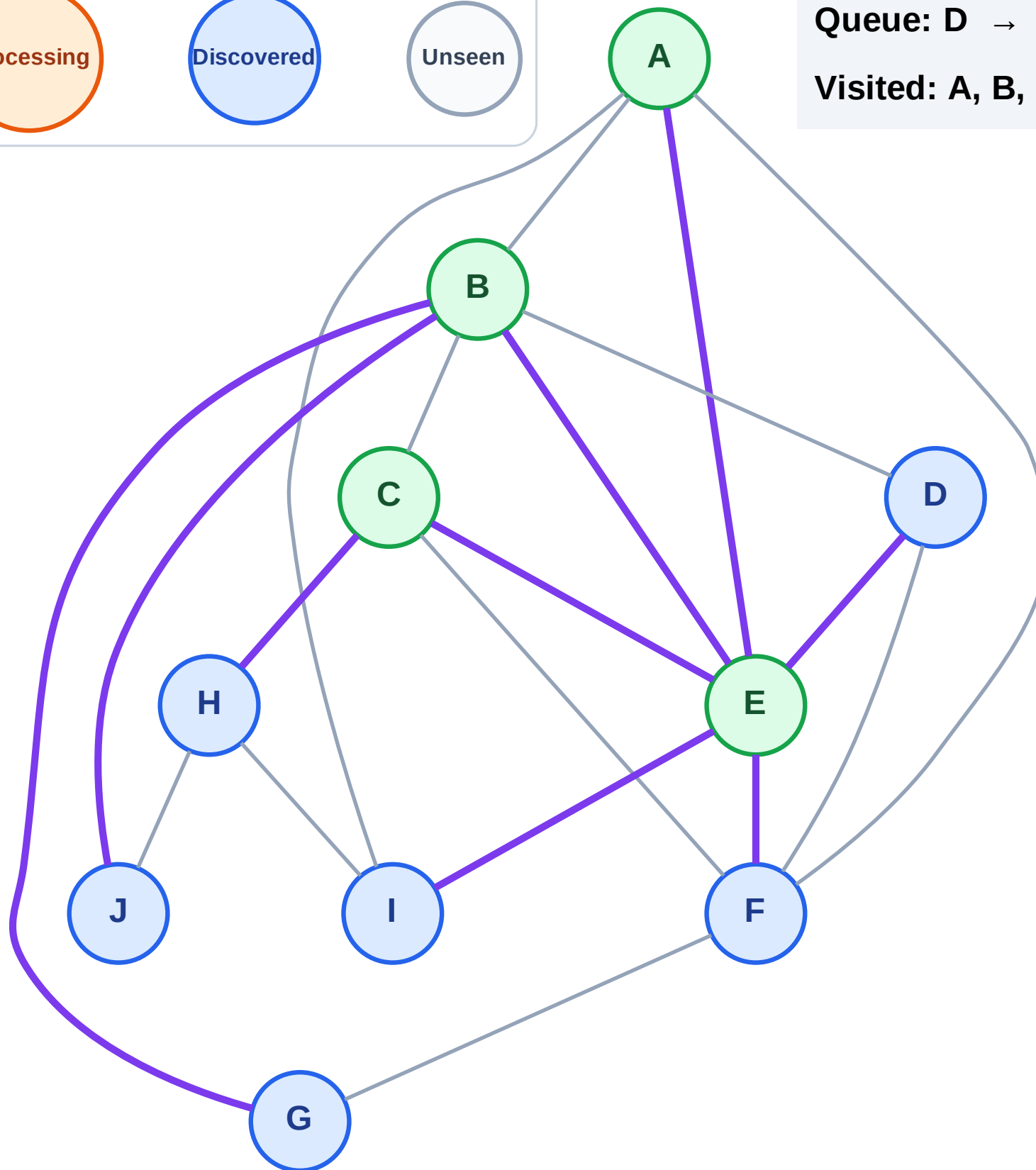
Discovered



Unseen

Queue: D → F → I → G → J → H

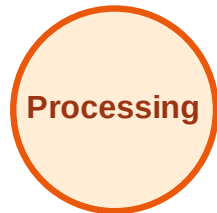
Visited: A, B, C, E



BFS Traversal

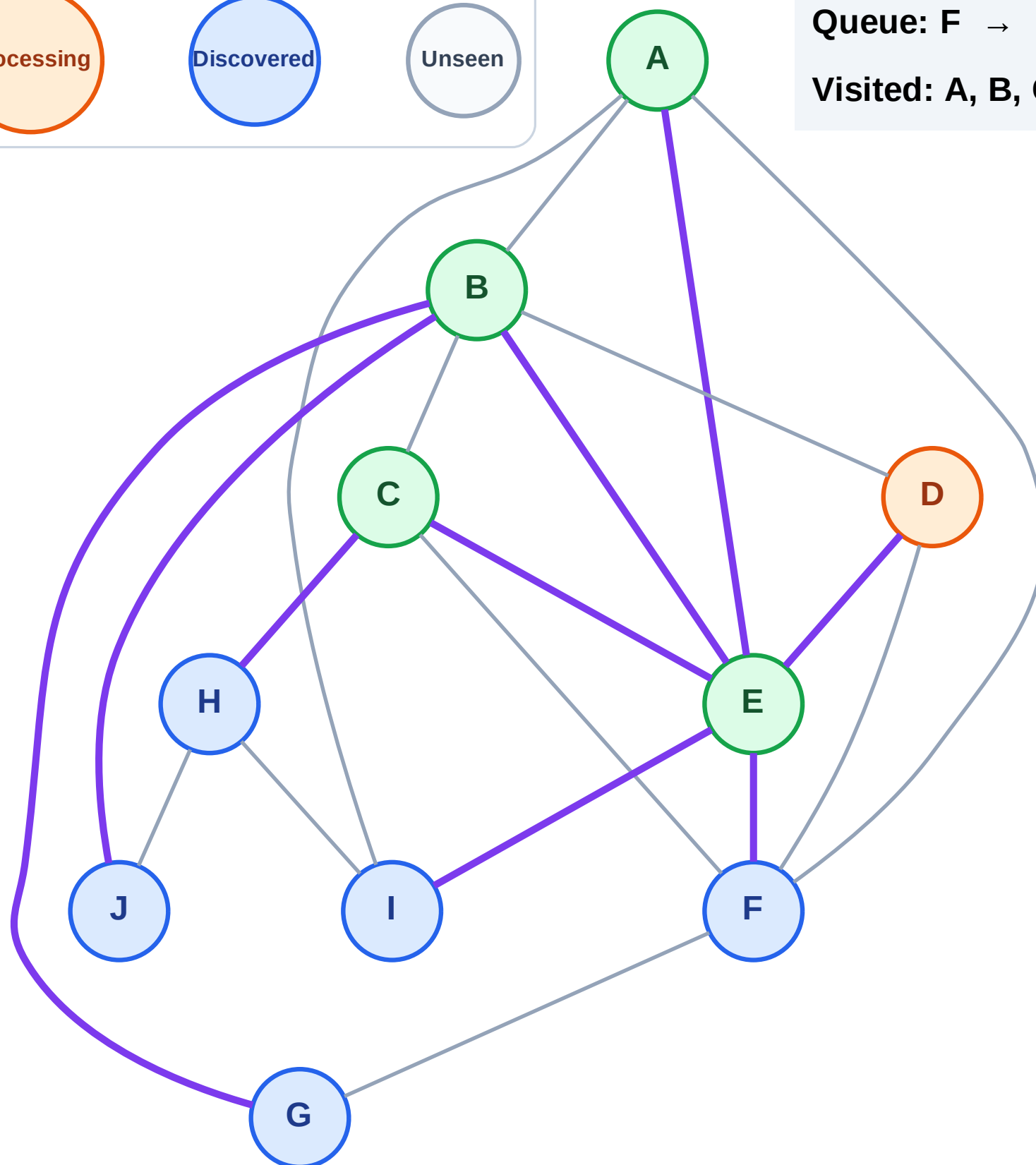
Step 10: Process node D

Legend



Queue: F → I → G → J → H

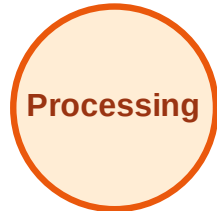
Visited: A, B, C, E



BFS Traversal

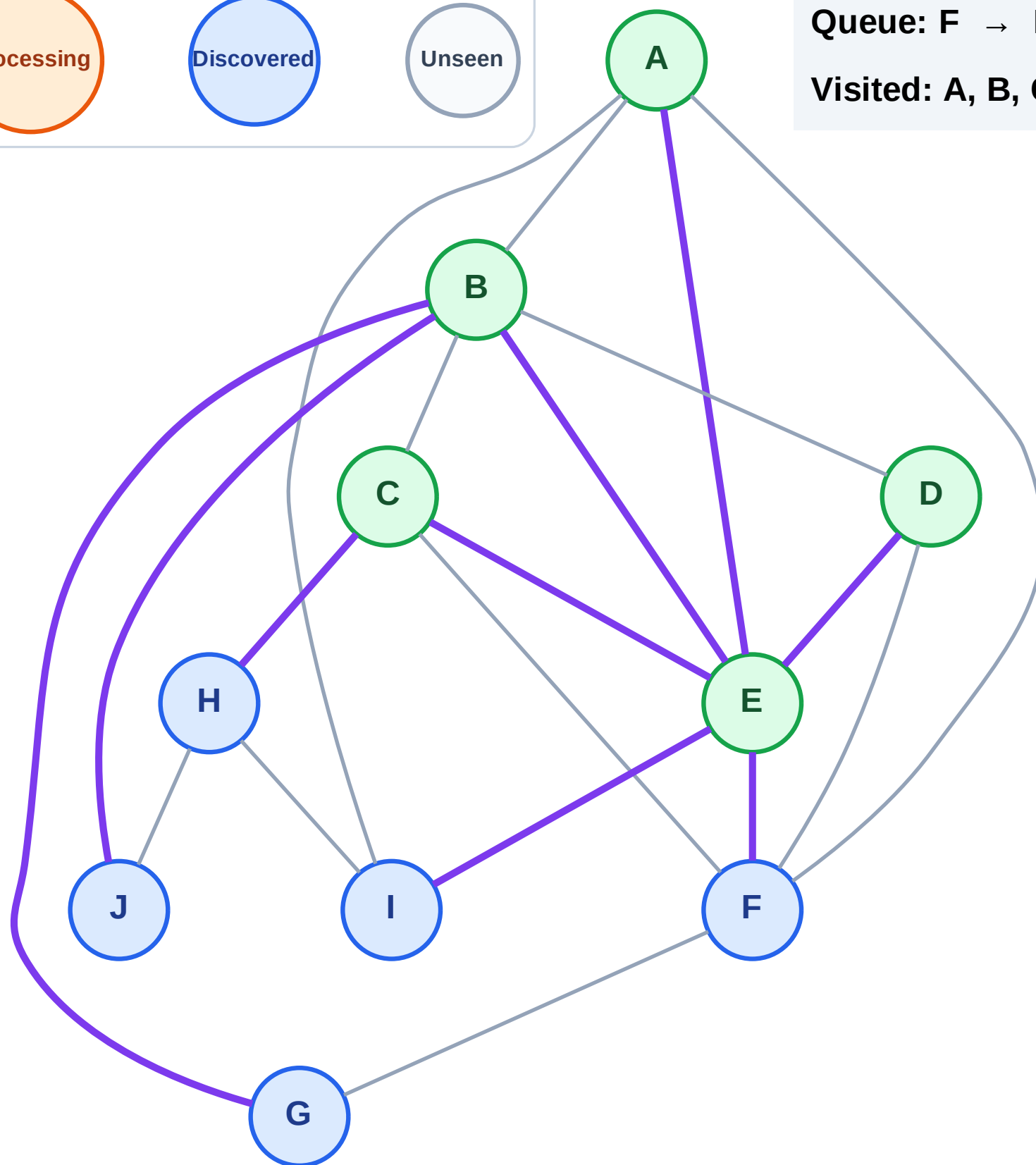
Step 11: No new neighbors from D

Legend



Queue: F → I → G → J → H

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node F

Legend

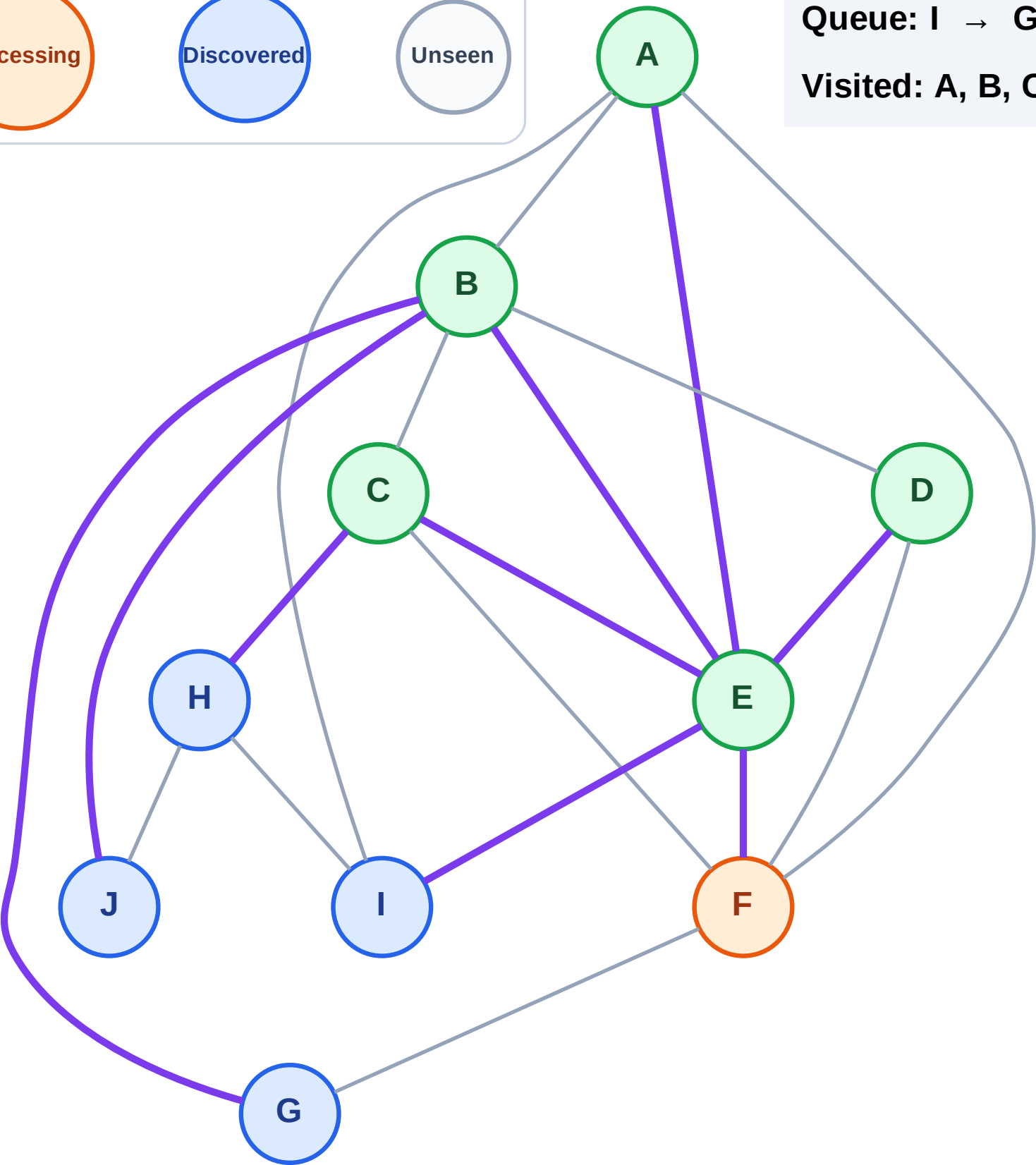
Complete

Processing

Discovered

Unseen

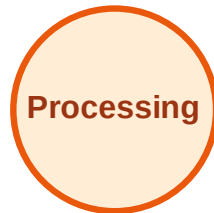
Queue: I → G → J → H
Visited: A, B, C, D, E



BFS Traversal

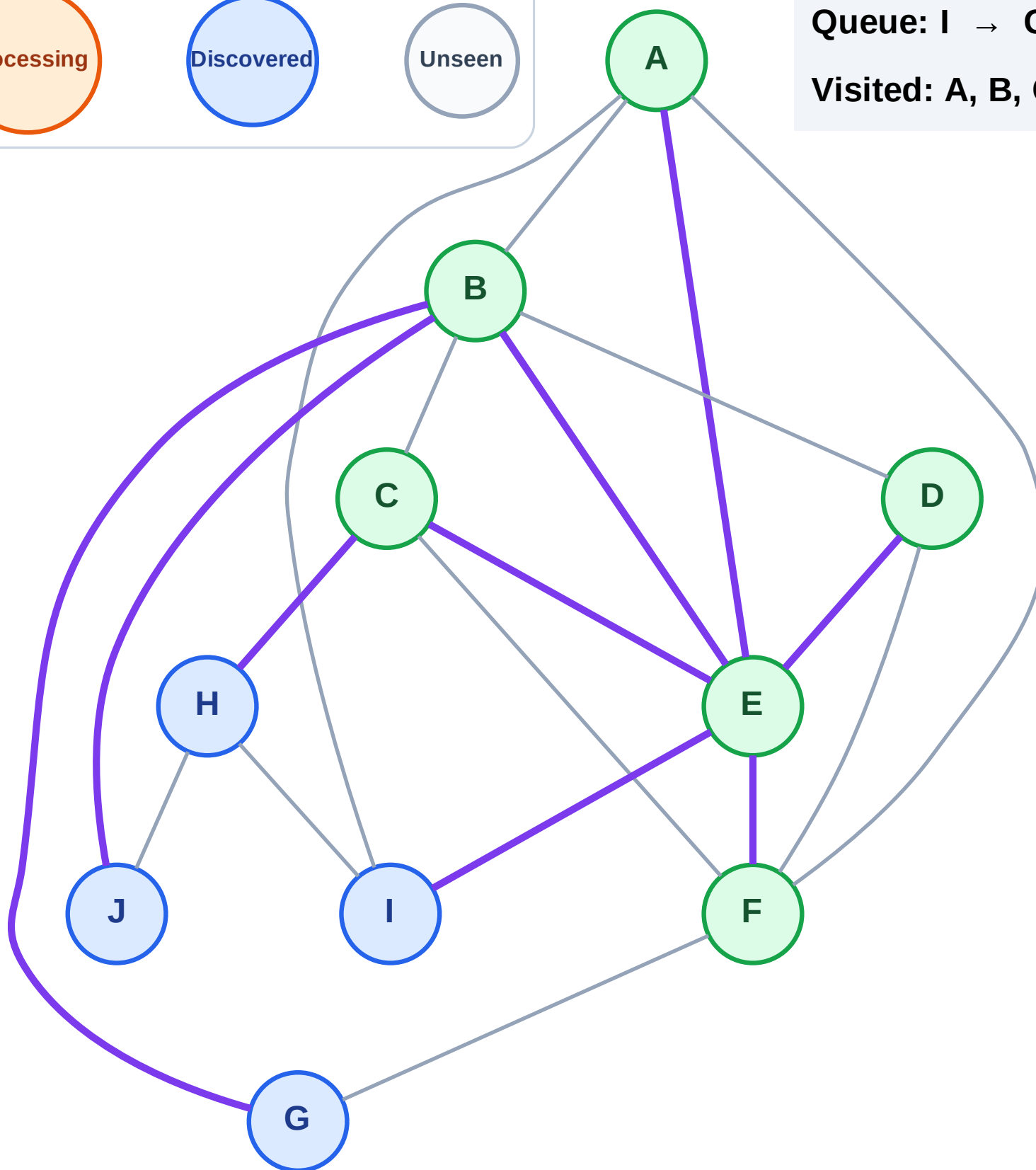
Step 13: No new neighbors from F

Legend



Queue: I → G → J → H

Visited: A, B, C, D, E, F



BFS Traversal

Step 14: Process node I

Legend

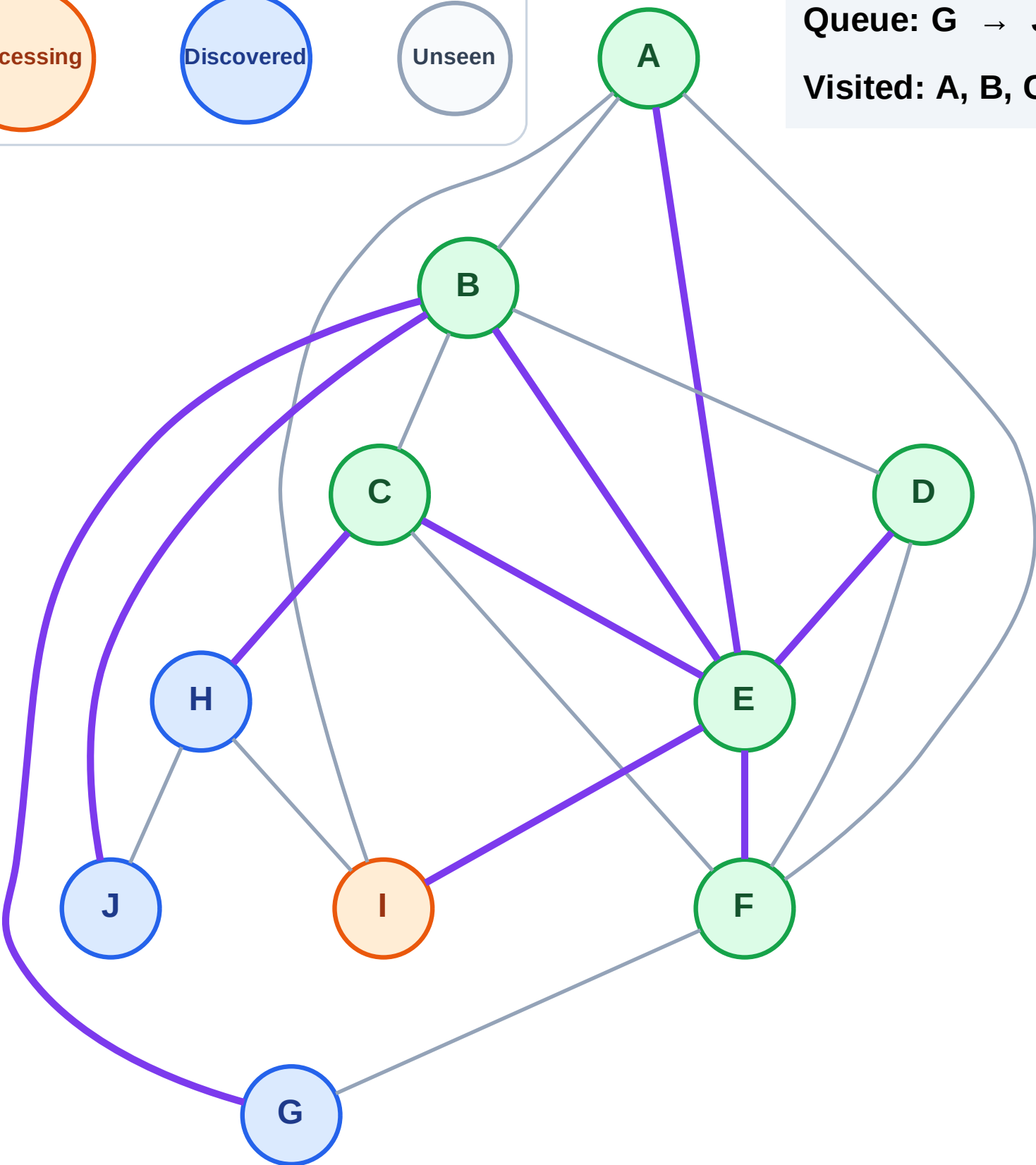
Complete

Processing

Discovered

Unseen

Queue: G → J → H
Visited: A, B, C, D, E, F



BFS Traversal

Step 15: No new neighbors from I

Legend

Complete

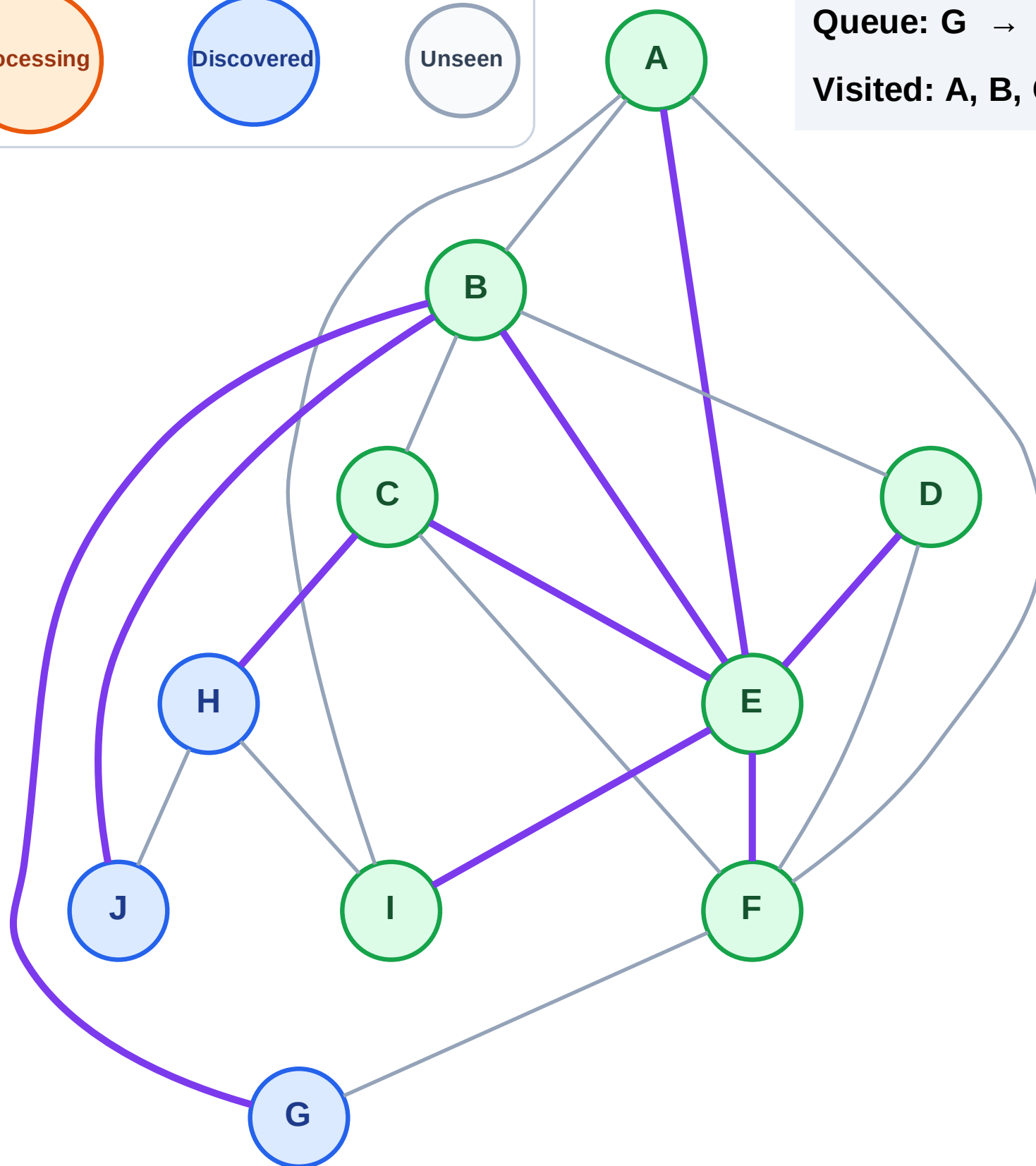
Processing

Discovered

Unseen

Queue: G → J → H

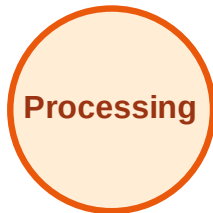
Visited: A, B, C, D, E, F, I



BFS Traversal

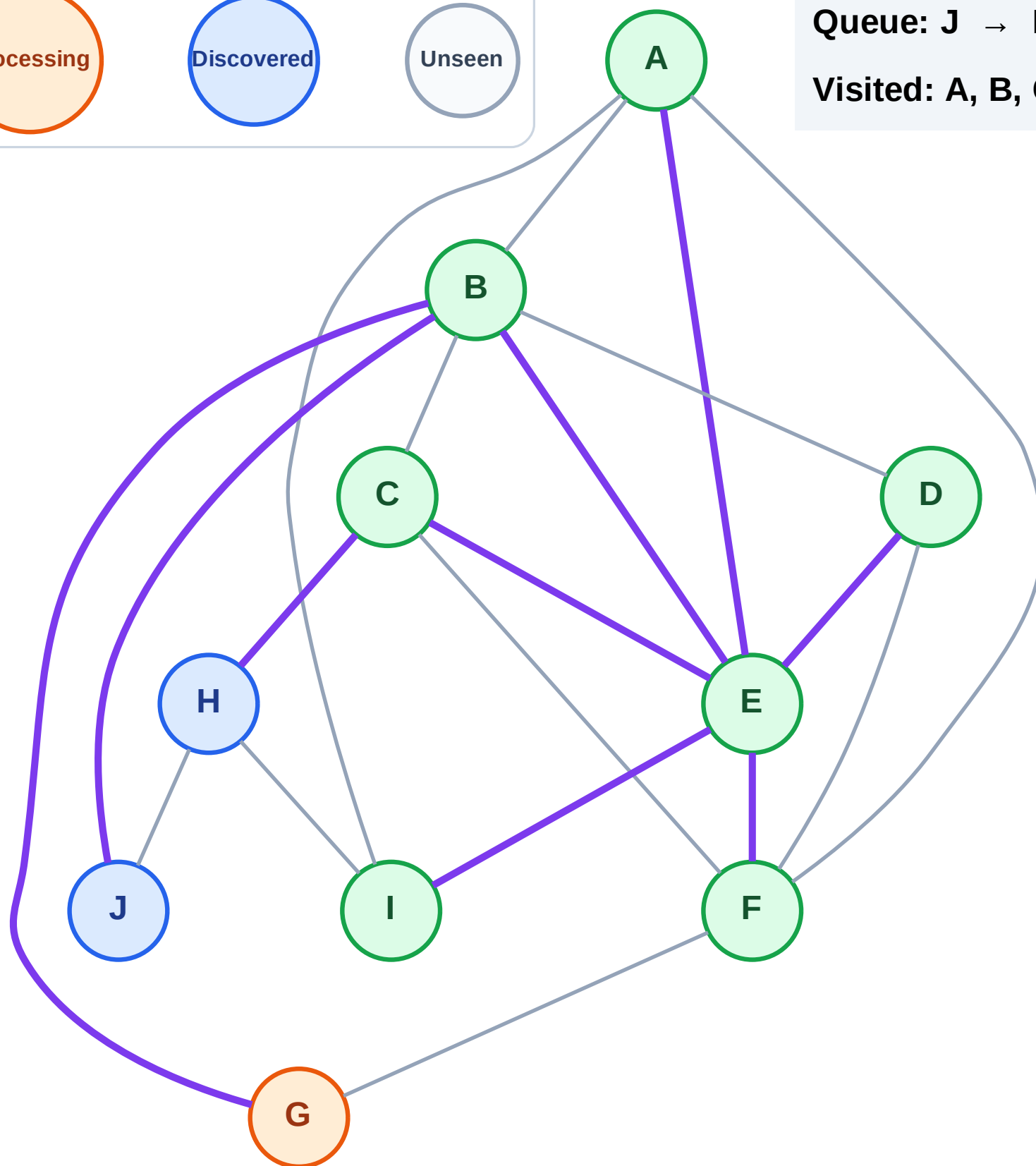
Step 16: Process node G

Legend



Queue: J → H

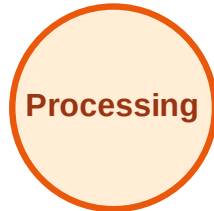
Visited: A, B, C, D, E, F, I



BFS Traversal

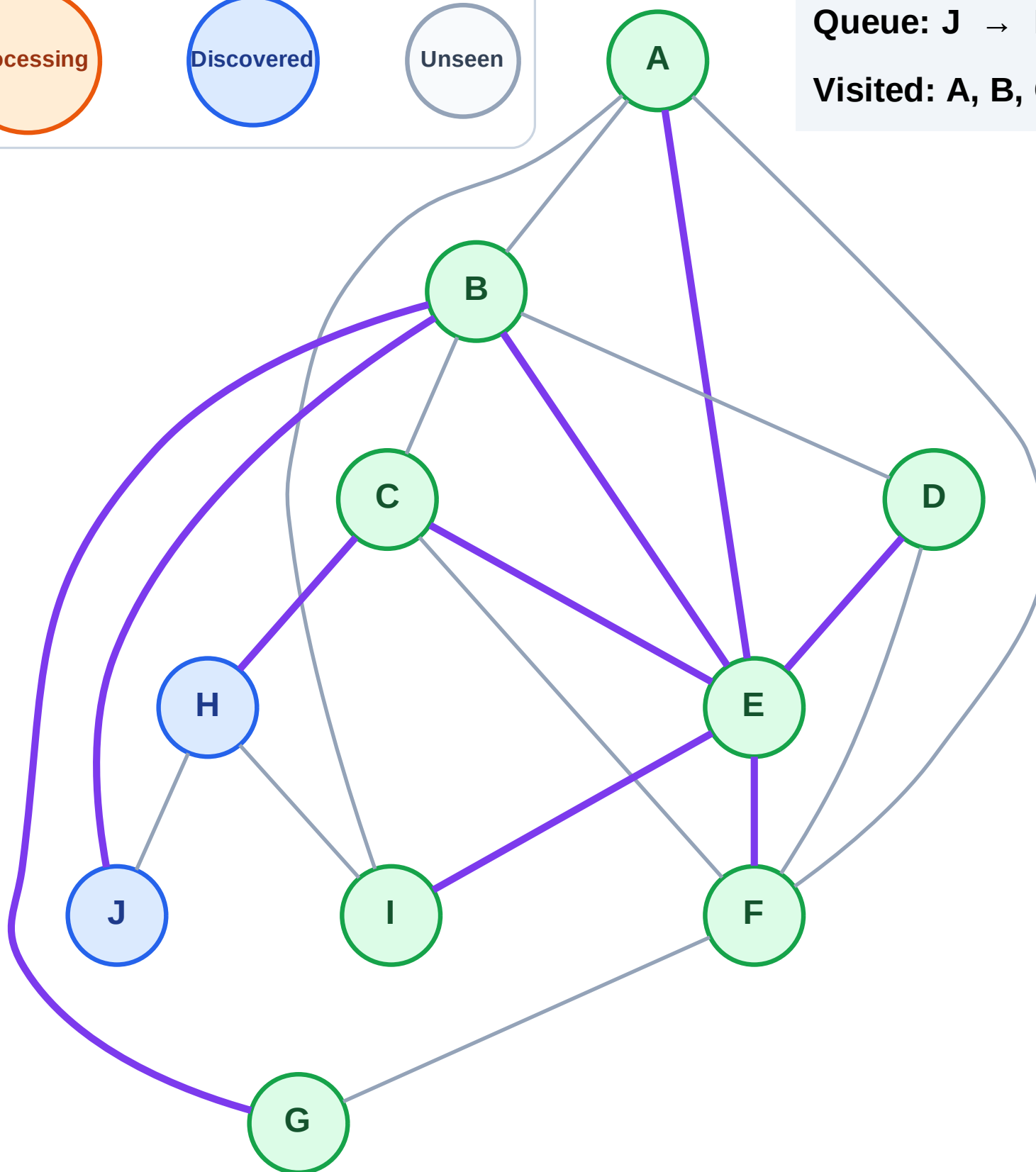
Step 17: No new neighbors from G

Legend



Queue: J → H

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node J

Legend

Complete

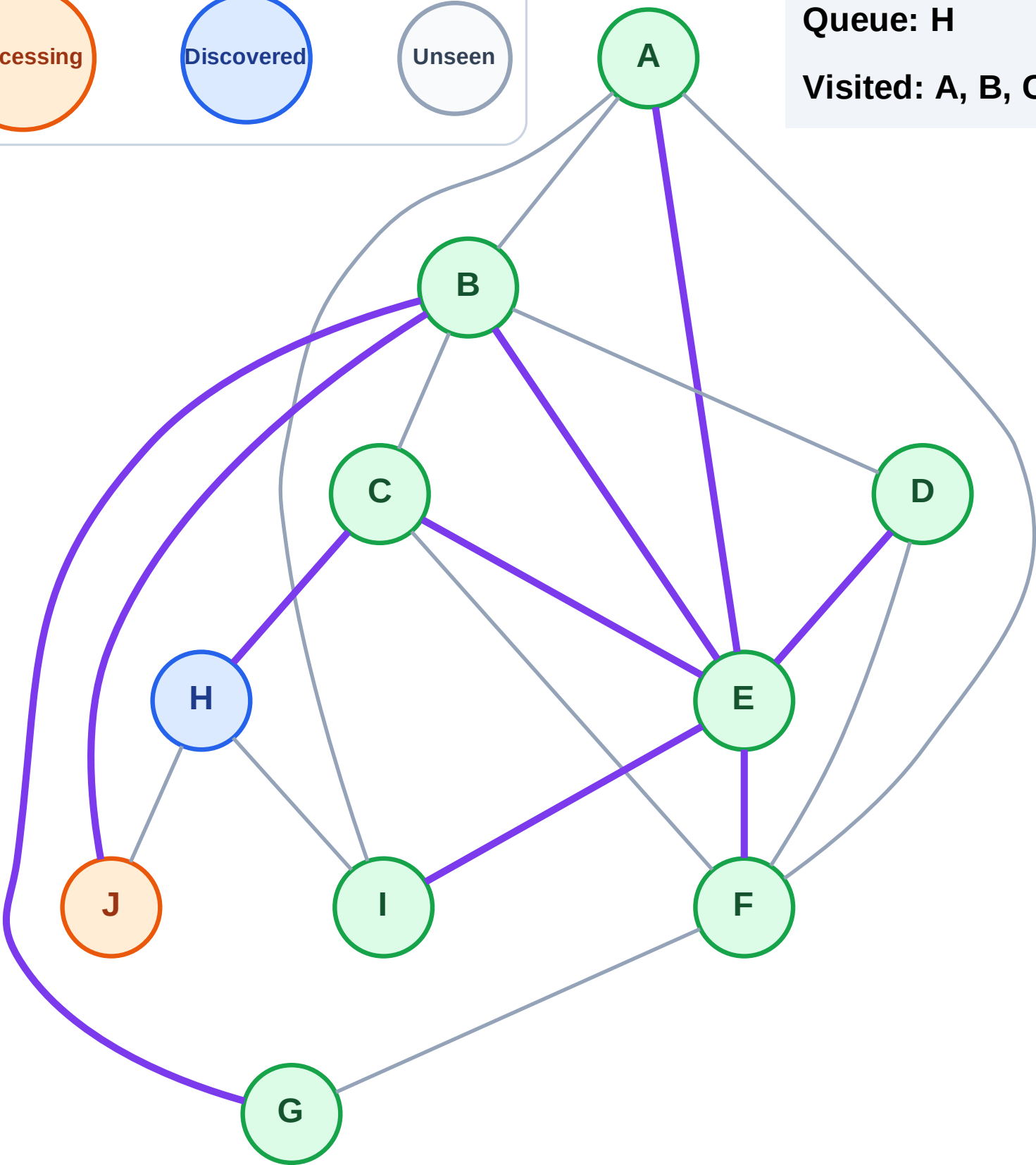
Processing

Discovered

Unseen

Queue: H

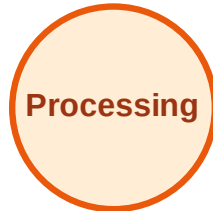
Visited: A, B, C, D, E, F, G, I



BFS Traversal

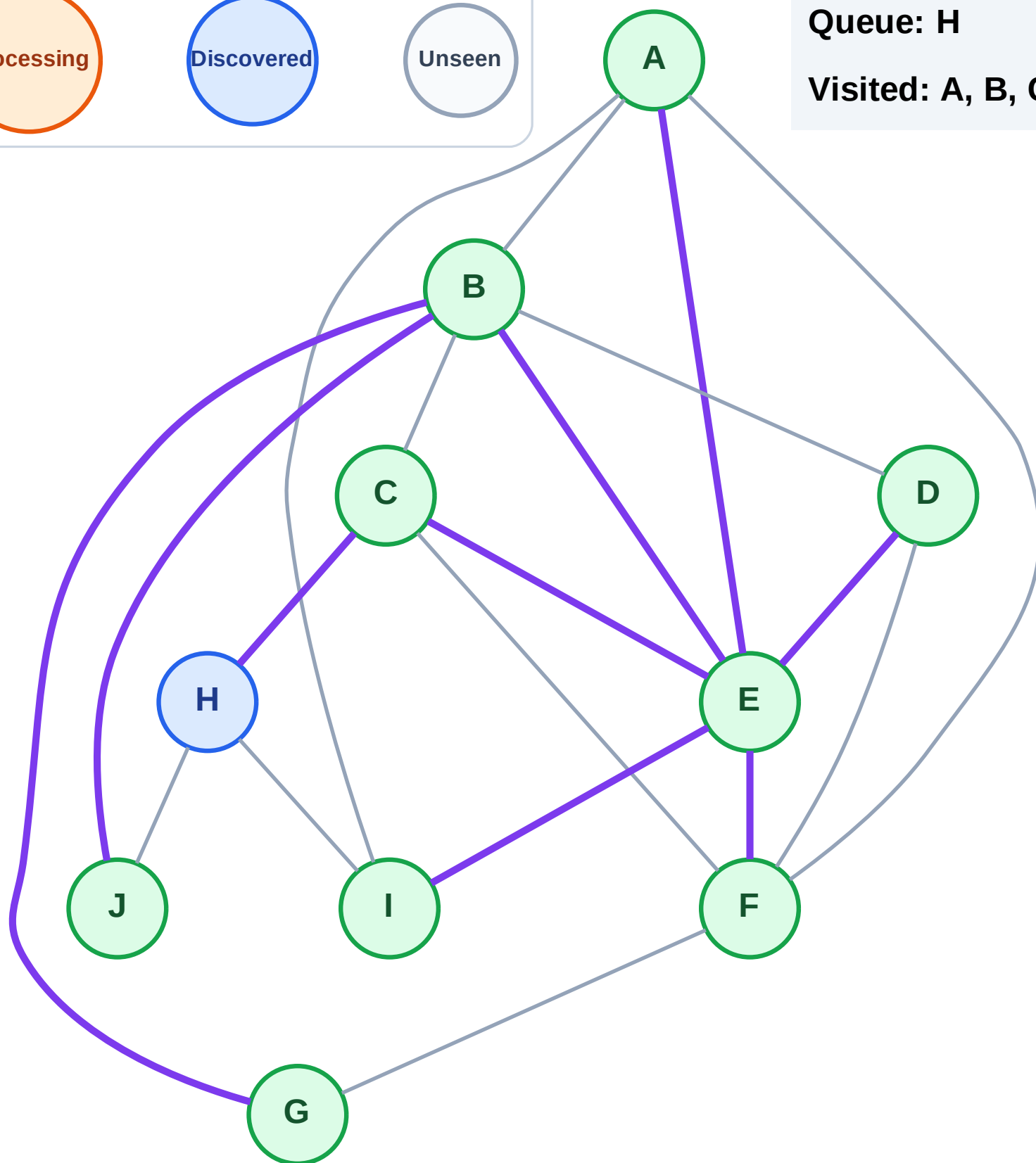
Step 19: No new neighbors from J

Legend



Queue: H

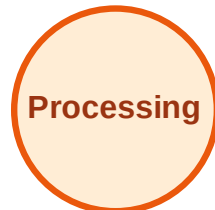
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

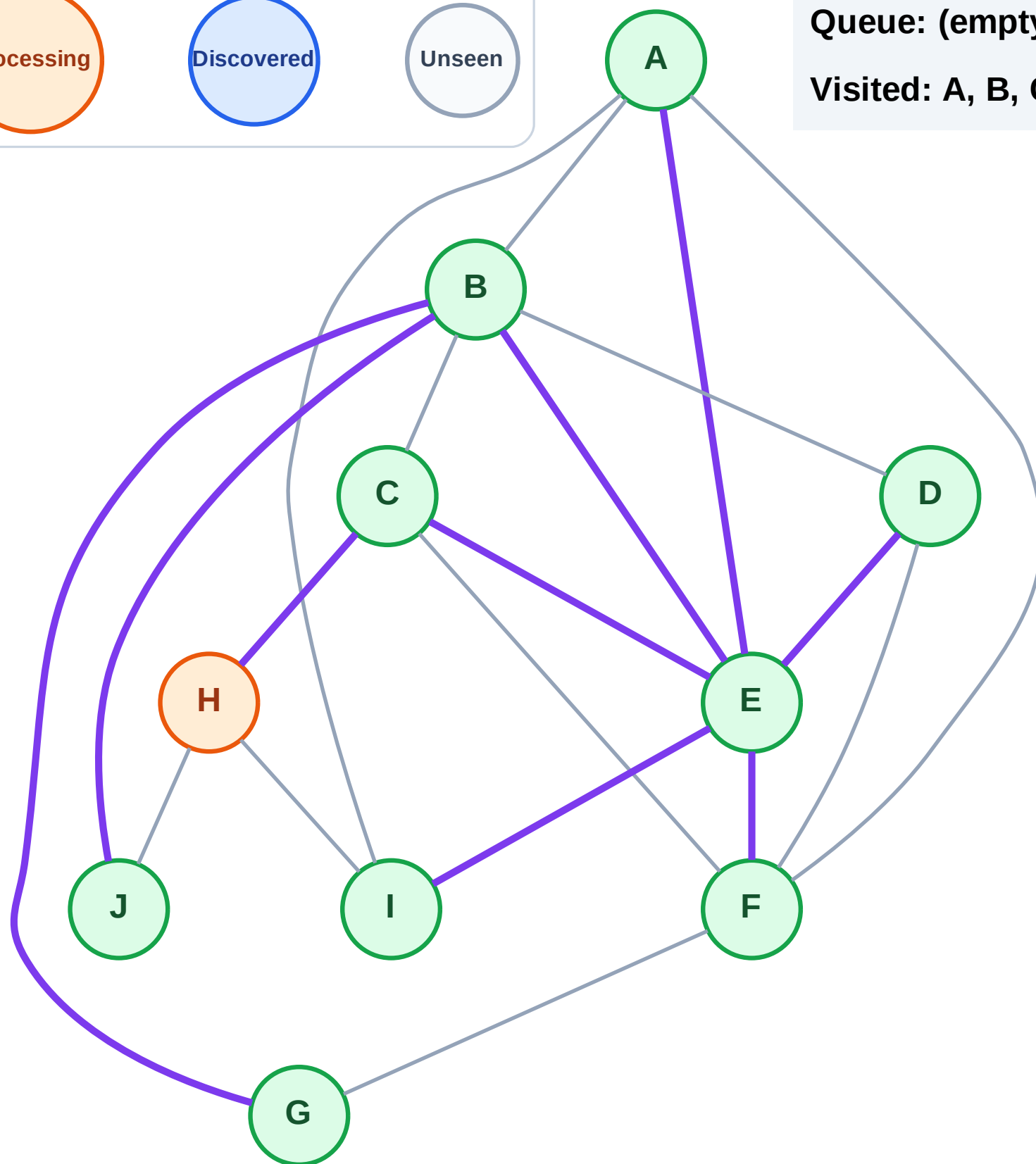
Step 20: Process node H

Legend



Queue: (empty)

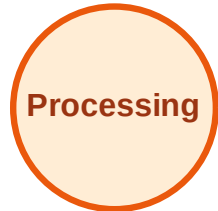
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

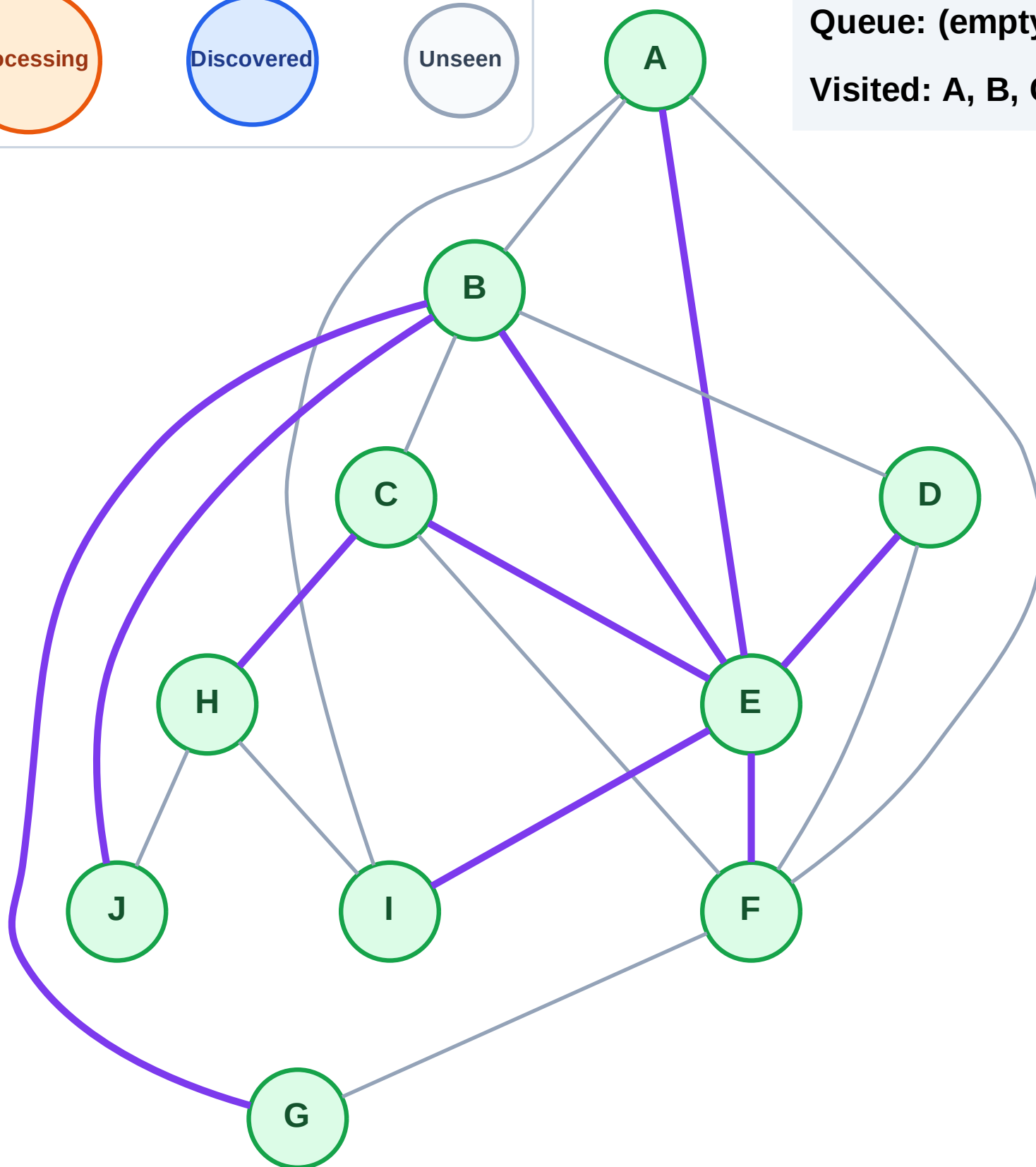
Step 21: No new neighbors from H

Legend



Queue: (empty)

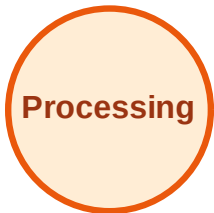
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

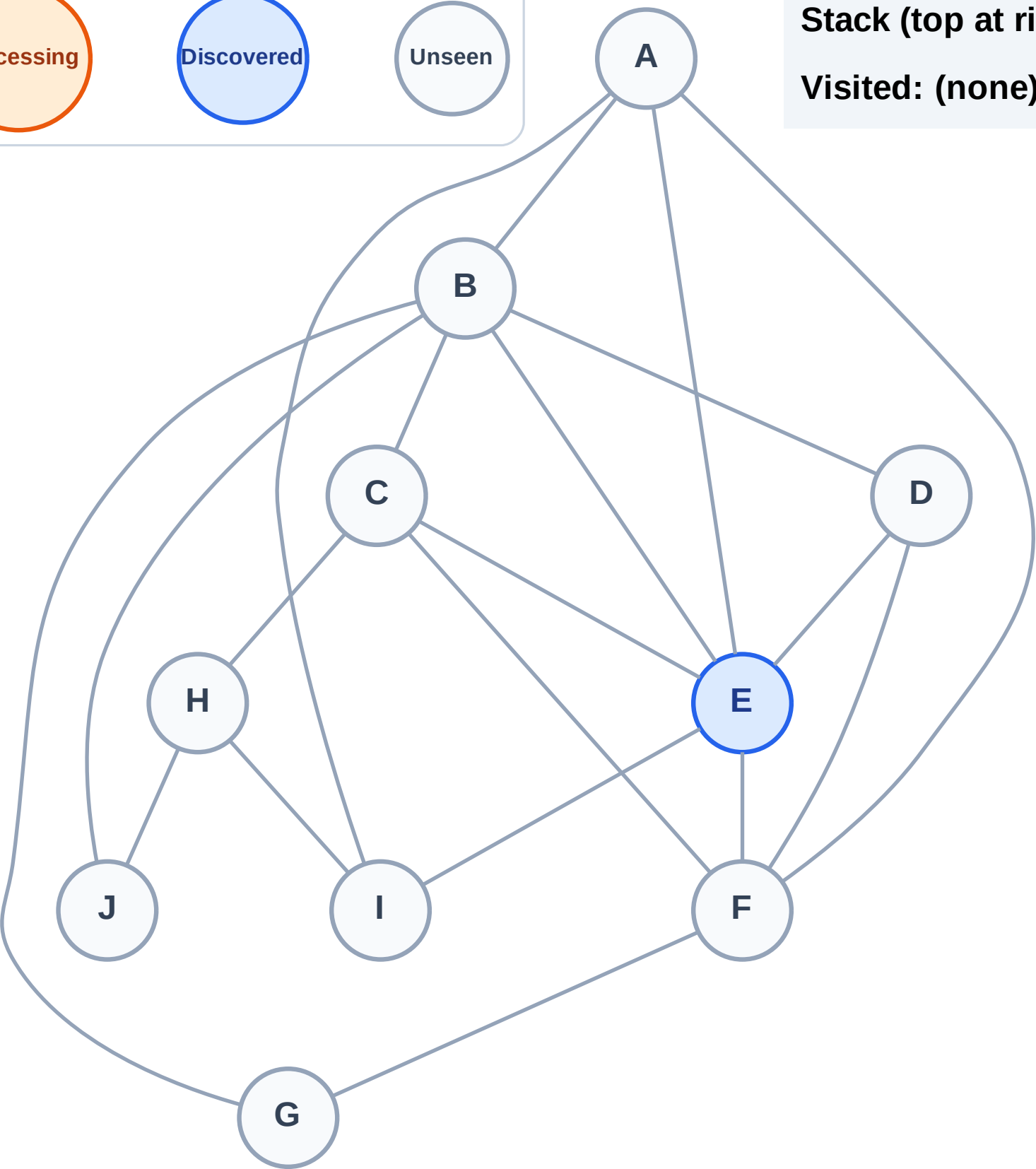
Step 1: Start at E

Legend



Stack (top at right): E

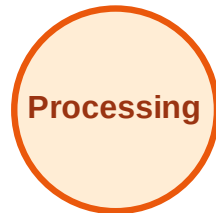
Visited: (none)



DFS Traversal

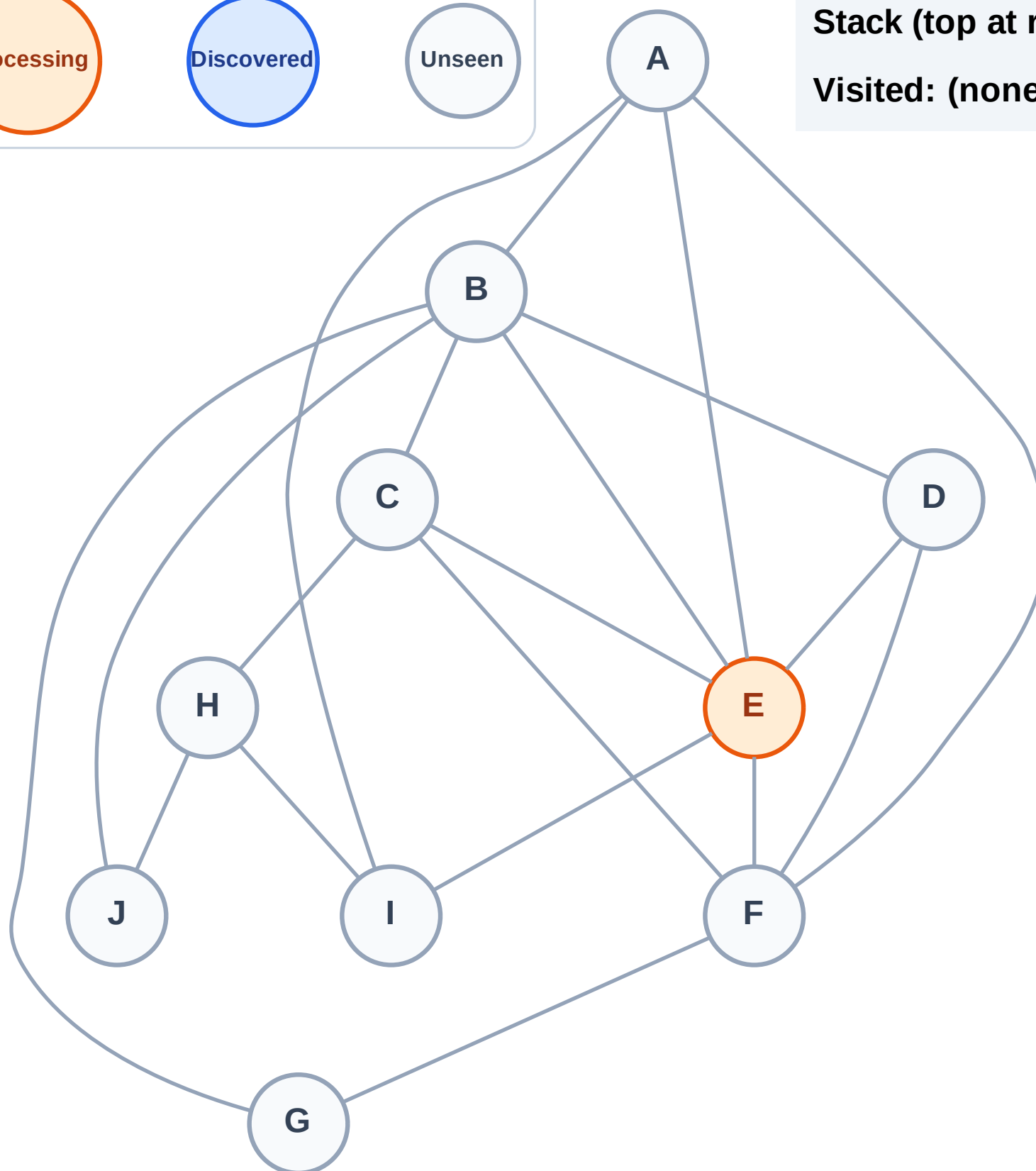
Step 2: Process node E

Legend



Stack (top at right): (empty)

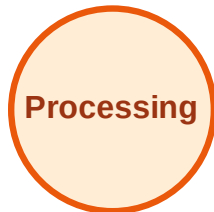
Visited: (none)



DFS Traversal

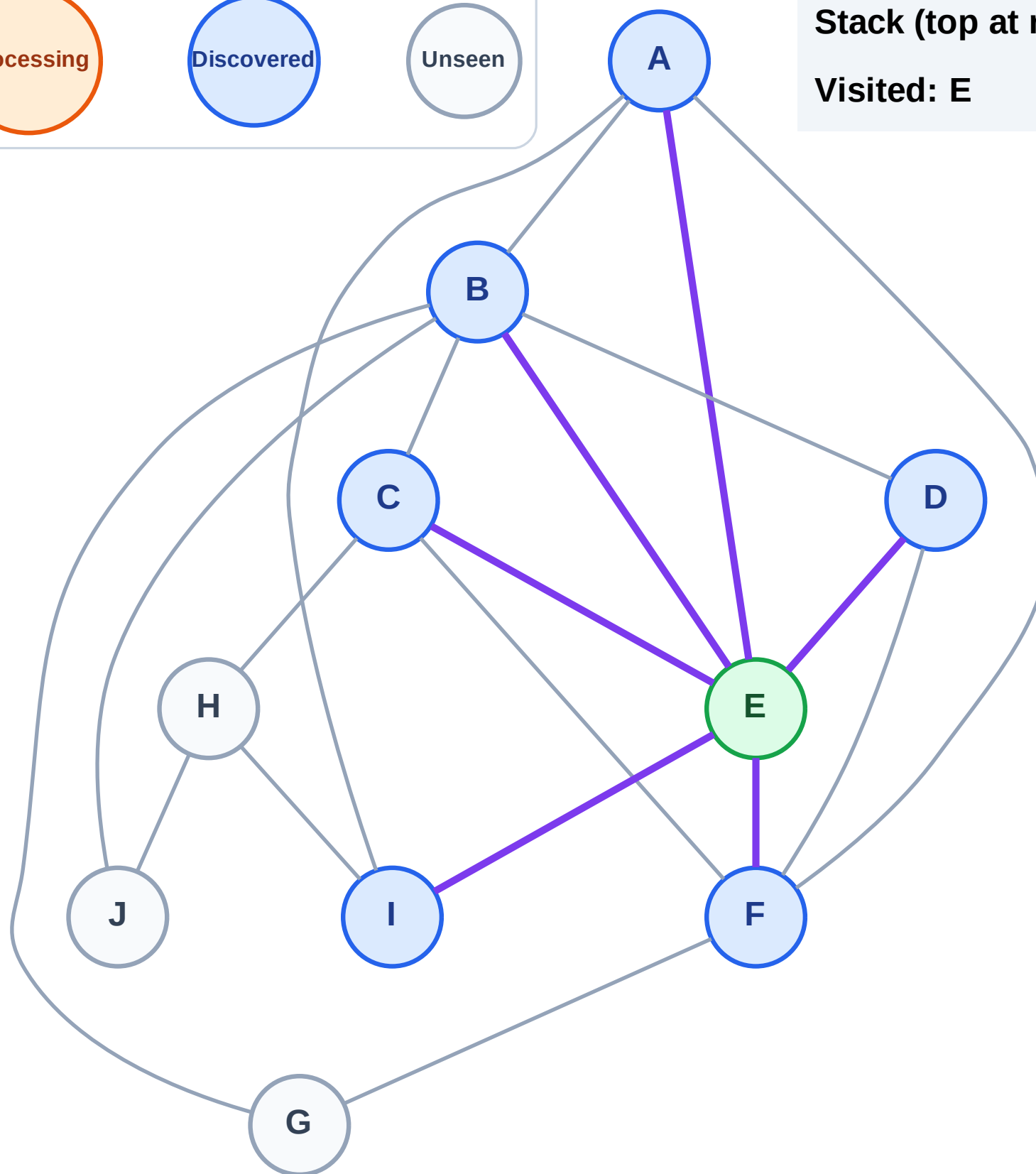
Step 3: Discover I, F, D, C, B, A from E

Legend



Stack (top at right): I | F | D | C | B | A

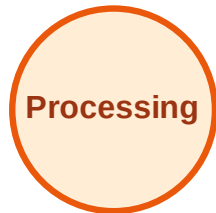
Visited: E



DFS Traversal

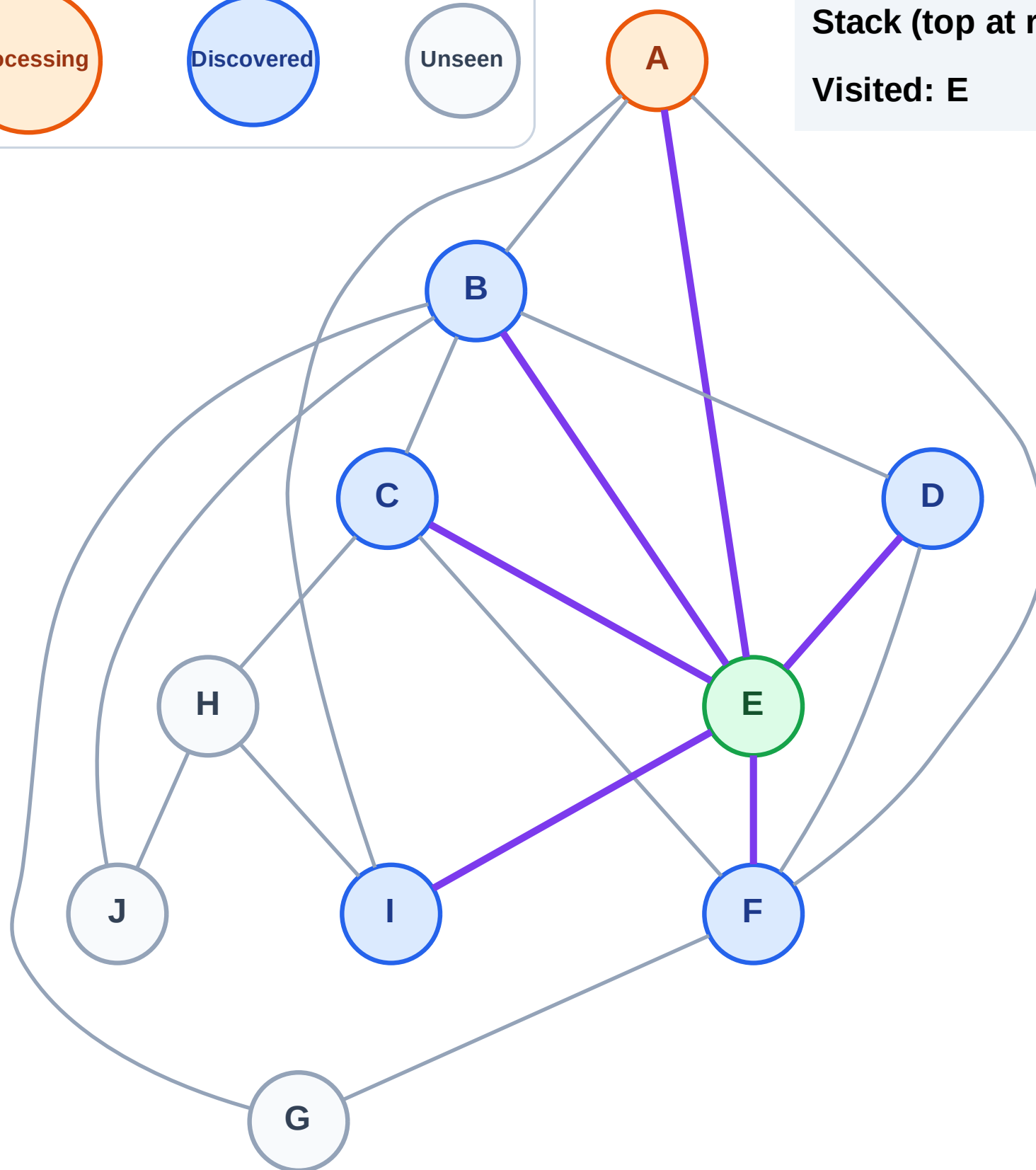
Step 4: Process node A

Legend



Stack (top at right): I | F | D | C | B

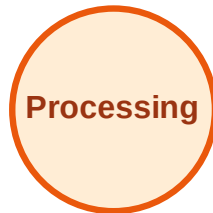
Visited: E



DFS Traversal

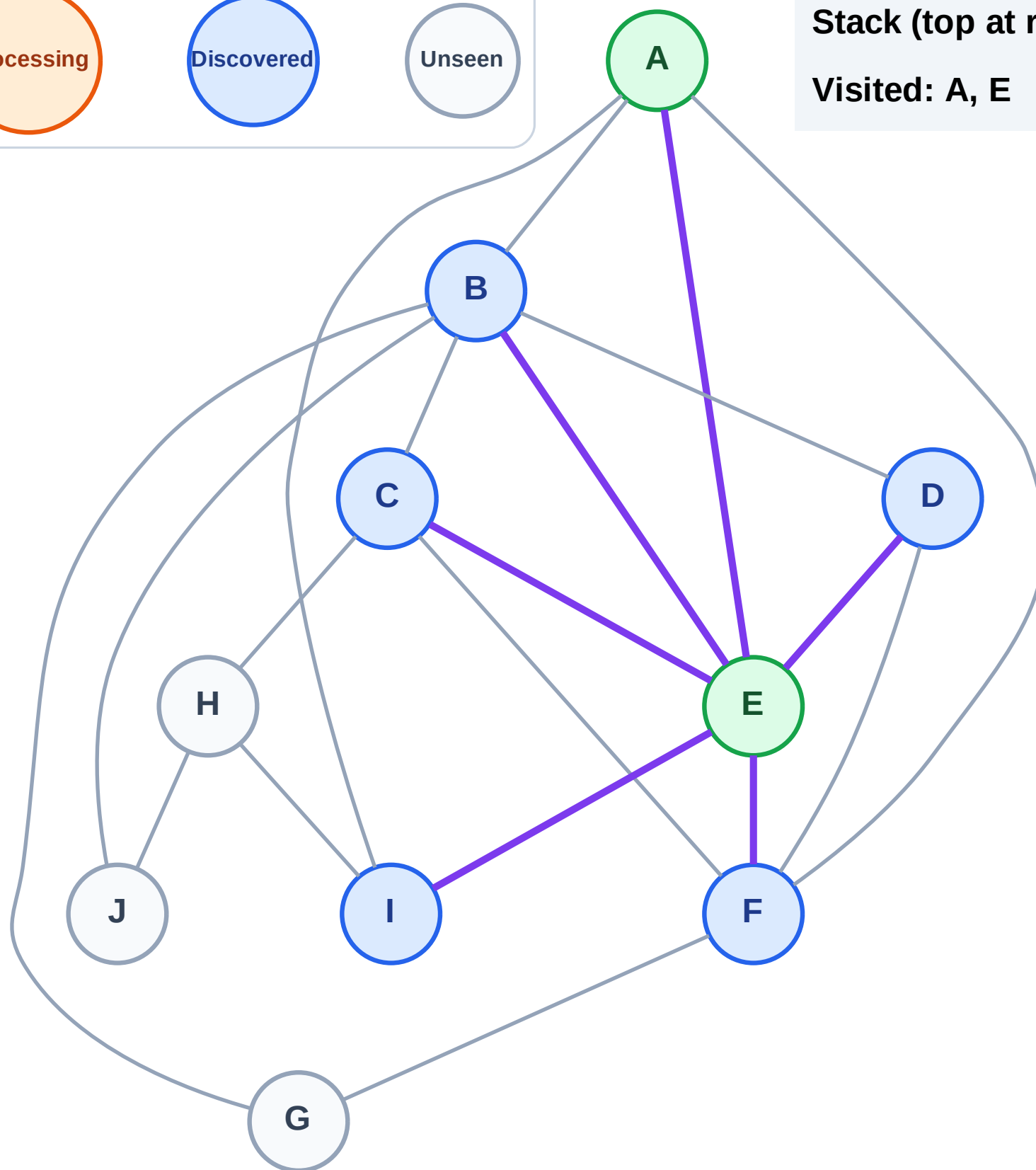
Step 5: No new neighbors from A

Legend



Stack (top at right): I | F | D | C | B

Visited: A, E



DFS Traversal

Step 6: Process node B

Legend

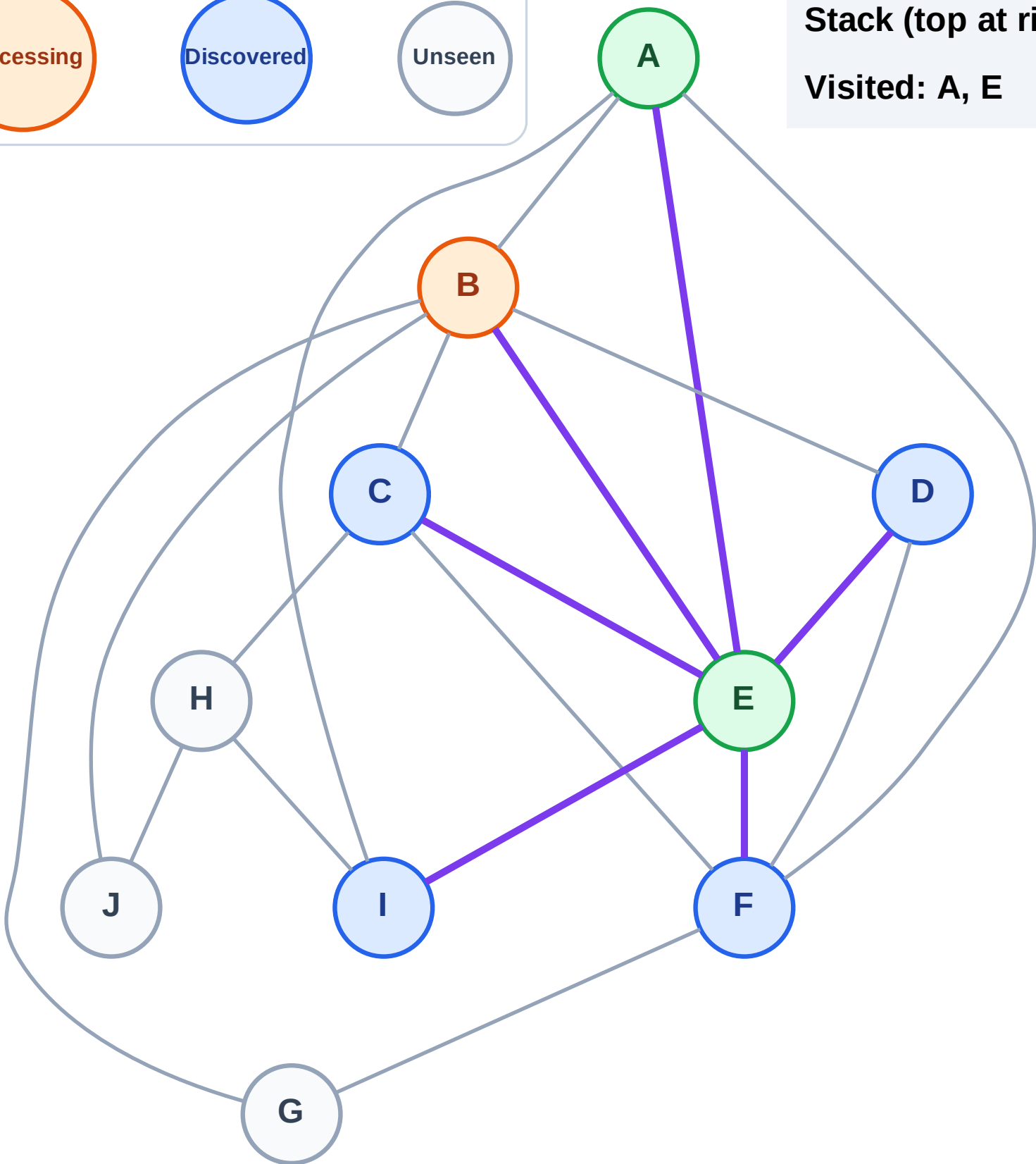
Complete

Processing

Discovered

Unseen

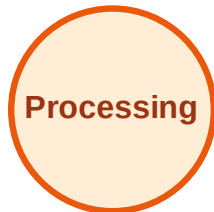
Stack (top at right): I | F | D | C
Visited: A, E



DFS Traversal

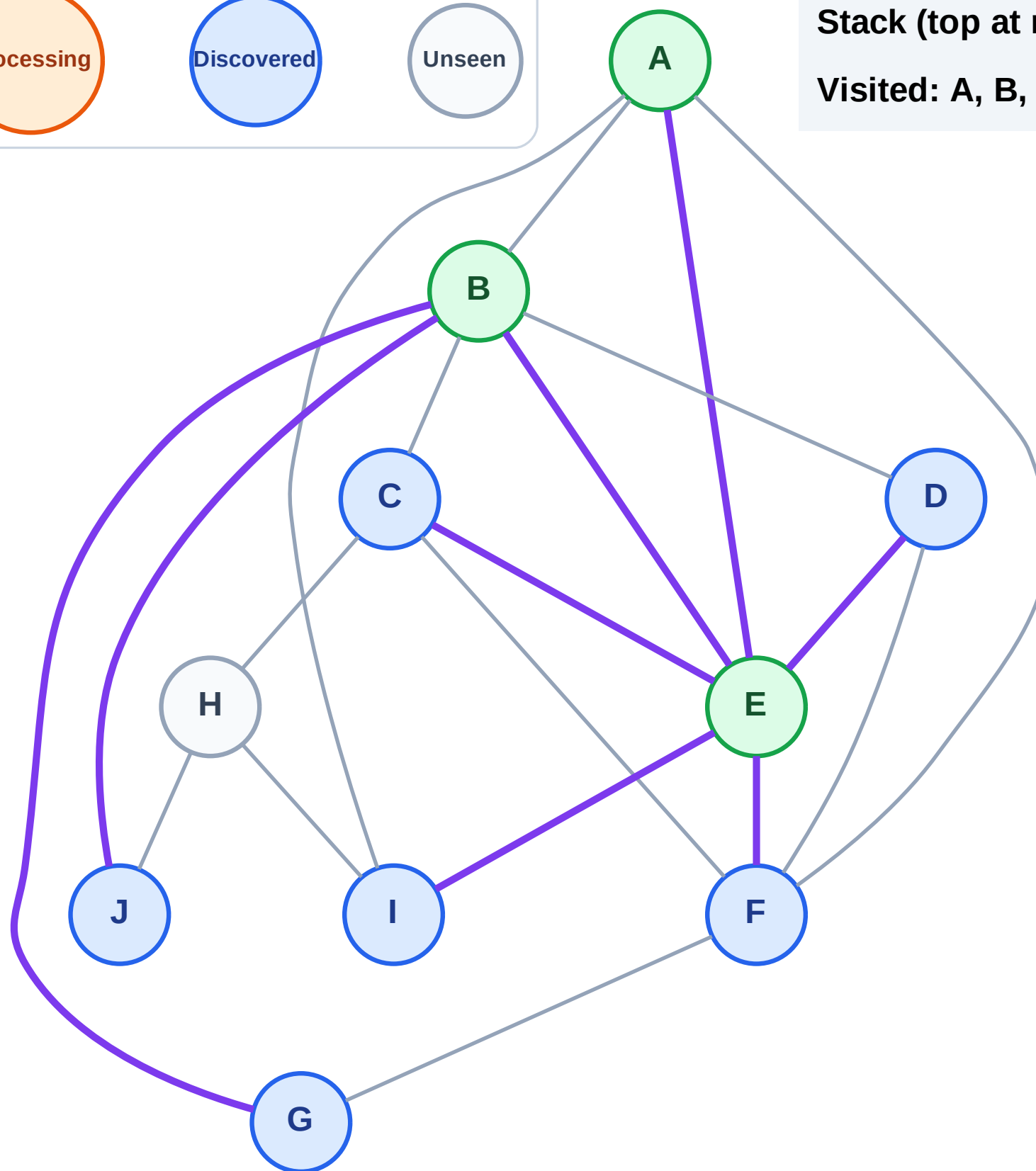
Step 7: Discover J, G from B

Legend



Stack (top at right): I | F | D | C | J | G

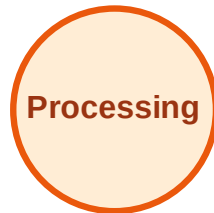
Visited: A, B, E



DFS Traversal

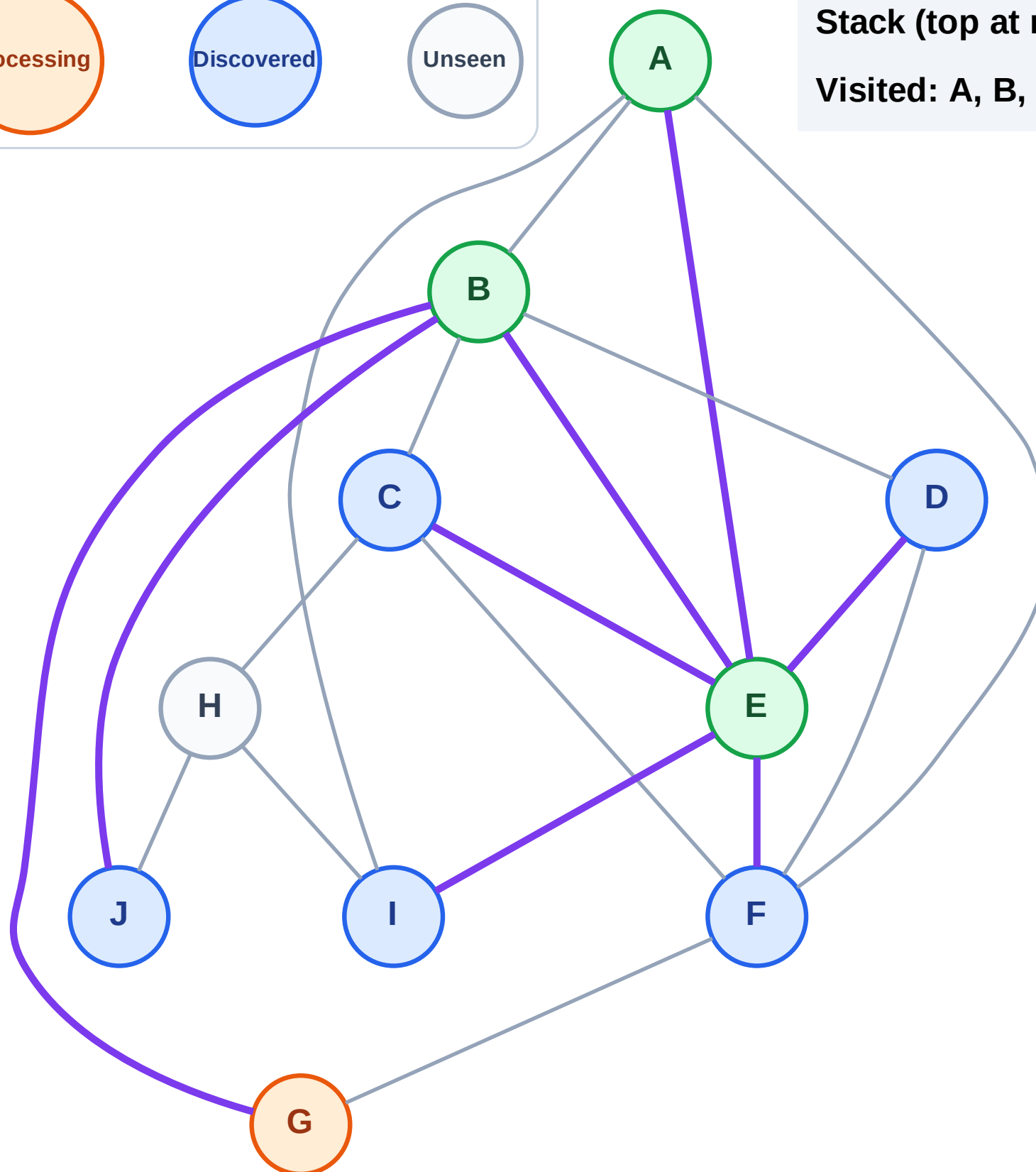
Step 8: Process node G

Legend



Stack (top at right): I | F | D | C | J

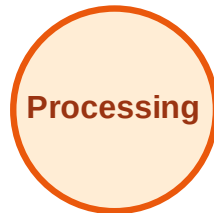
Visited: A, B, E



DFS Traversal

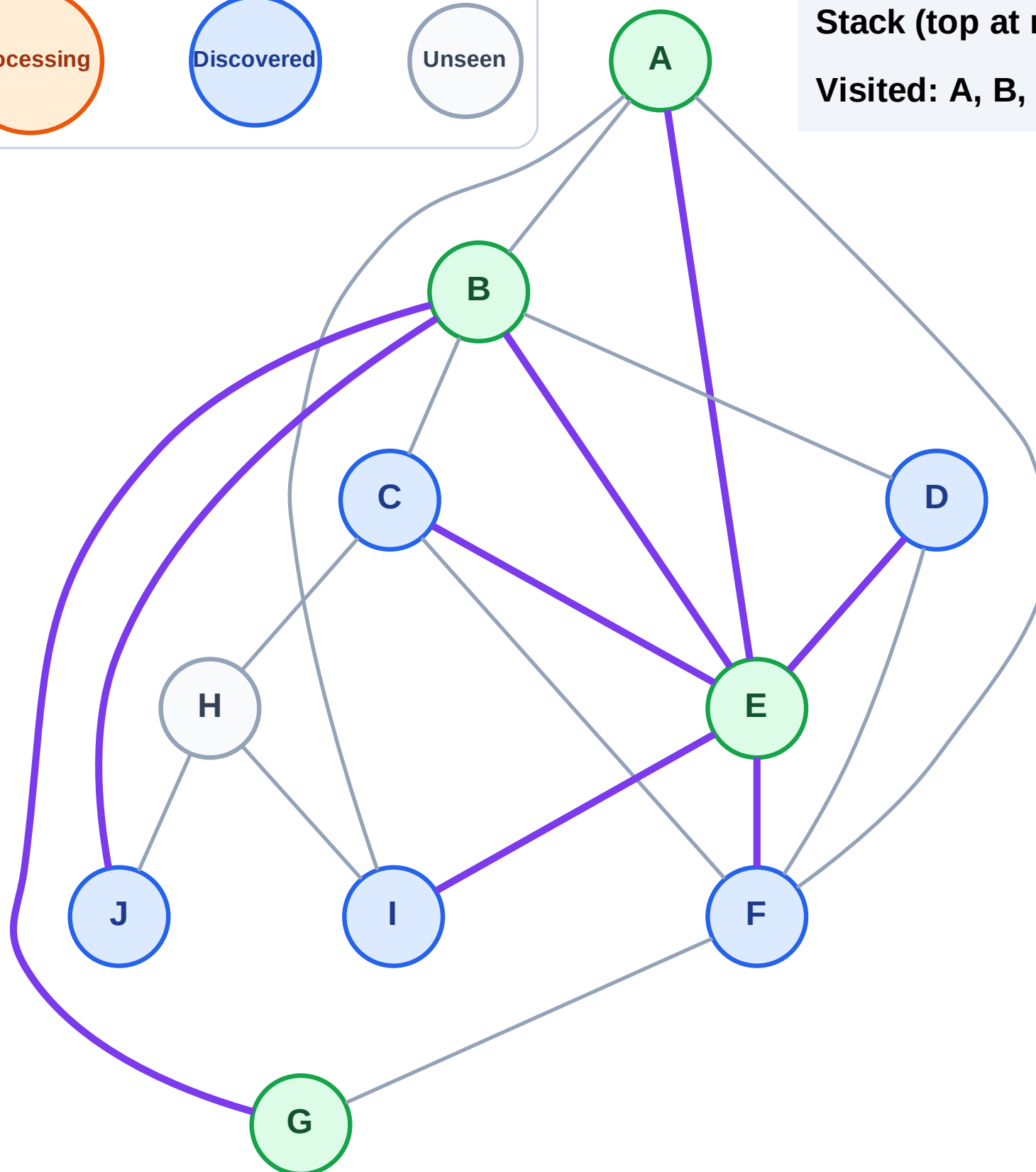
Step 9: No new neighbors from G

Legend



Stack (top at right): I | F | D | C | J

Visited: A, B, E, G



DFS Traversal

Step 10: Process node J

Legend

Complete

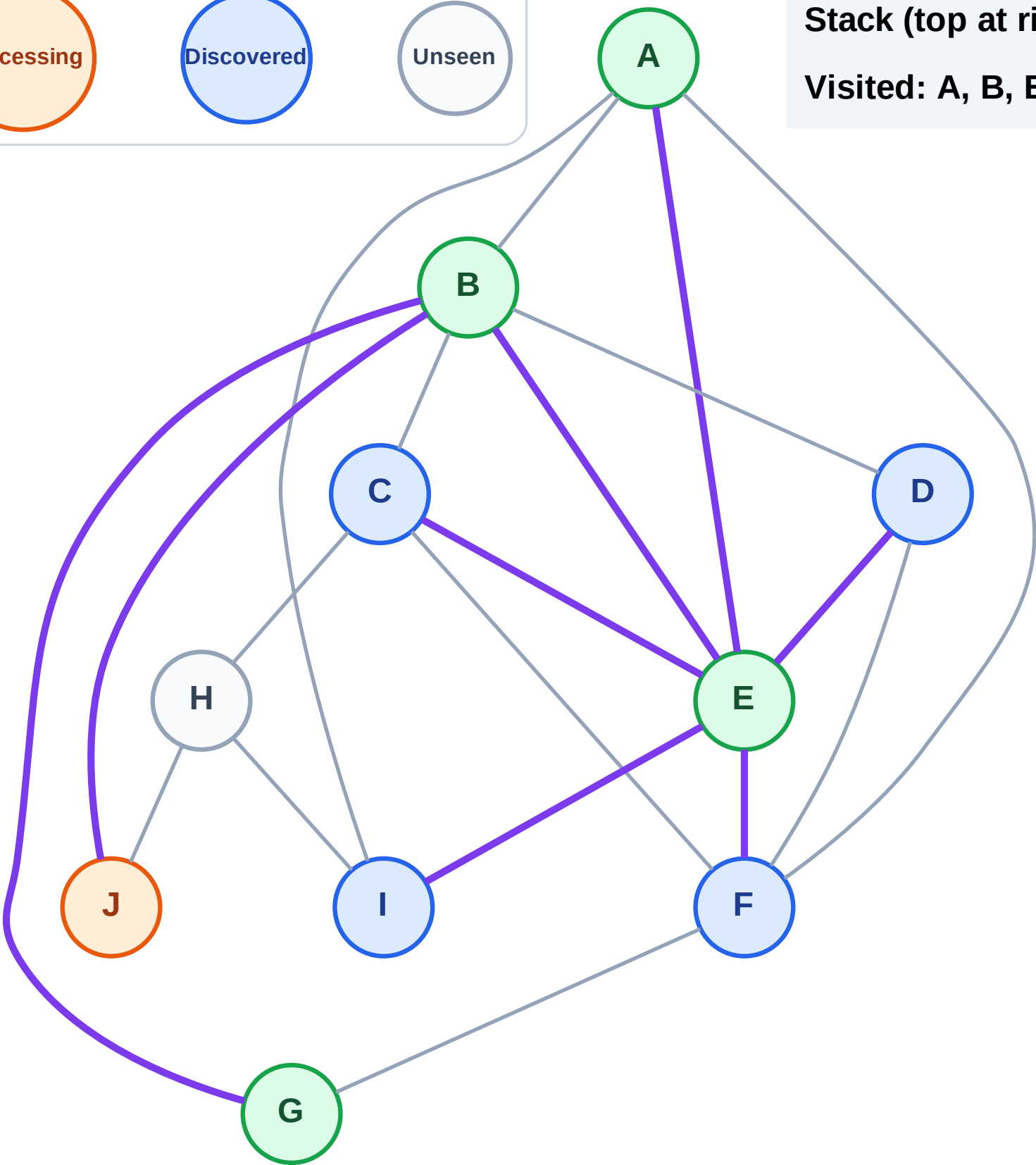
Processing

Discovered

Unseen

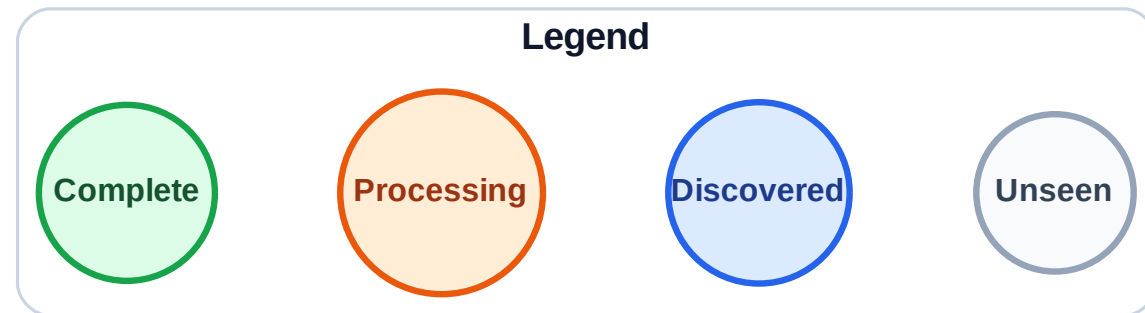
Stack (top at right): I | F | D | C

Visited: A, B, E, G



Step 11: Discover H from J

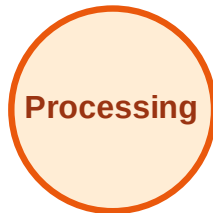
Step 11: Discover H from J



DFS Traversal

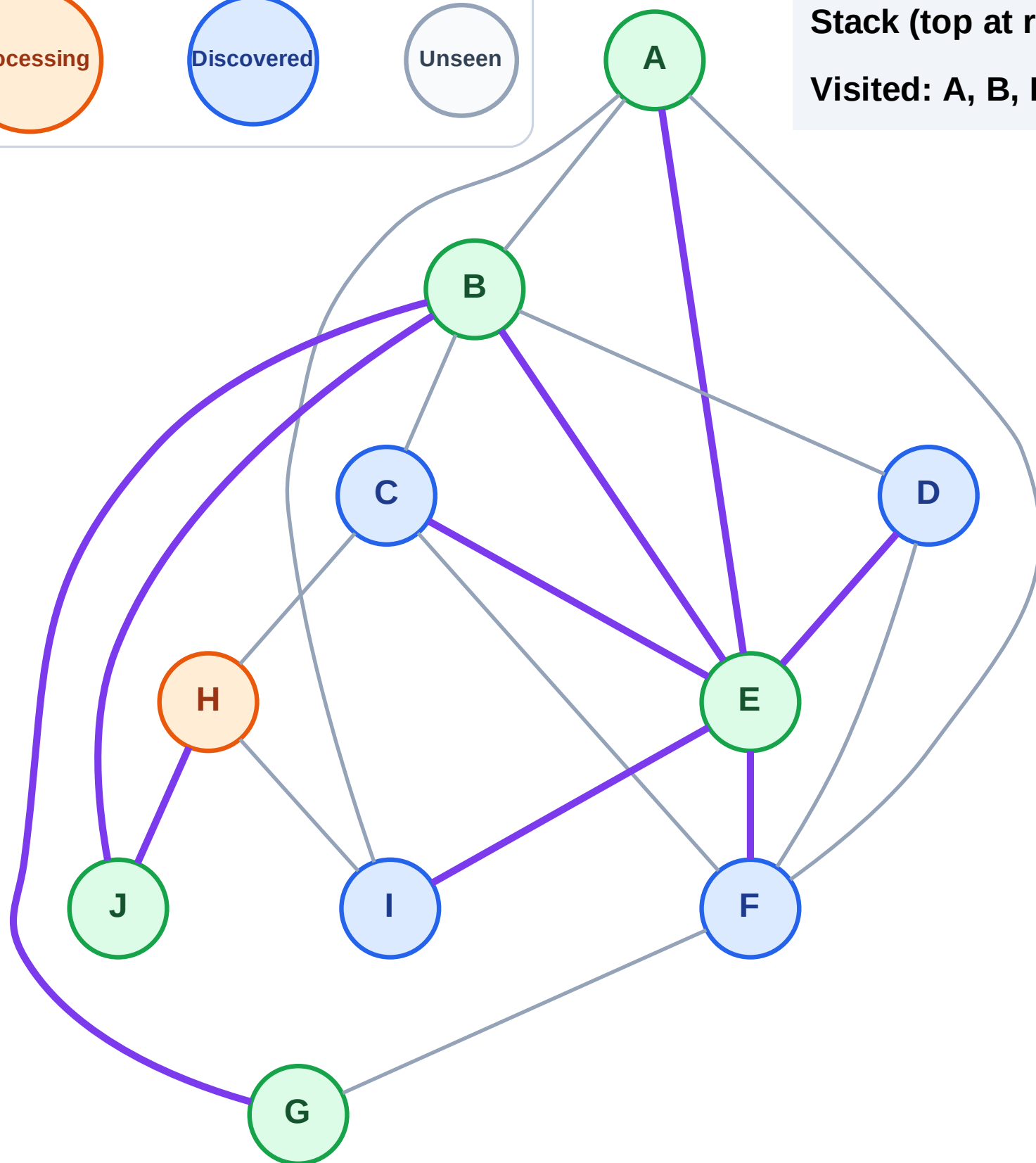
Step 12: Process node H

Legend



Stack (top at right): I | F | D | C

Visited: A, B, E, G, J



DFS Traversal

Step 13: No new neighbors from H

Legend

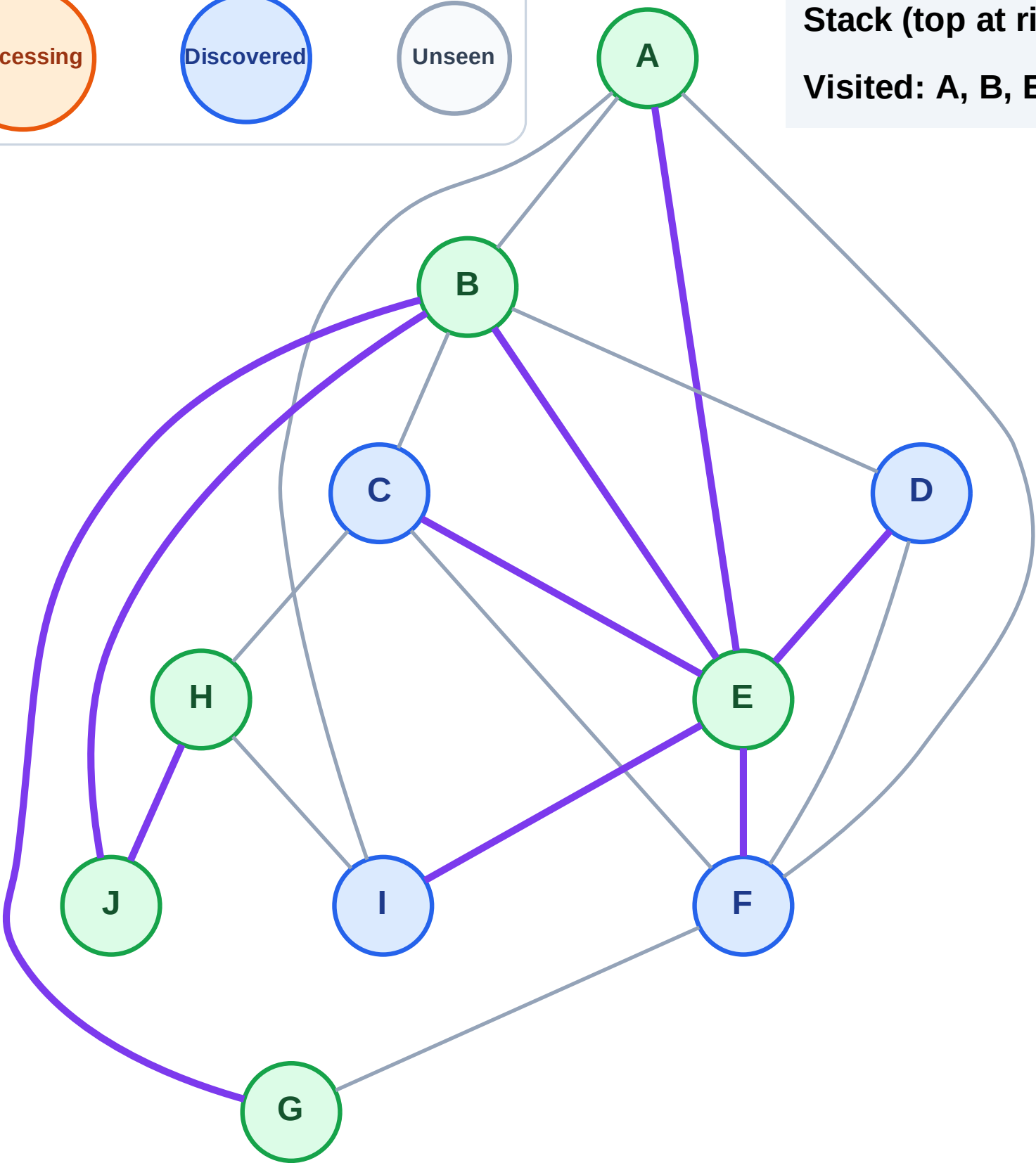
Complete

Processing

Discovered

Unseen

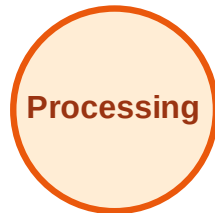
Stack (top at right): I | F | D | C
Visited: A, B, E, G, H, J



DFS Traversal

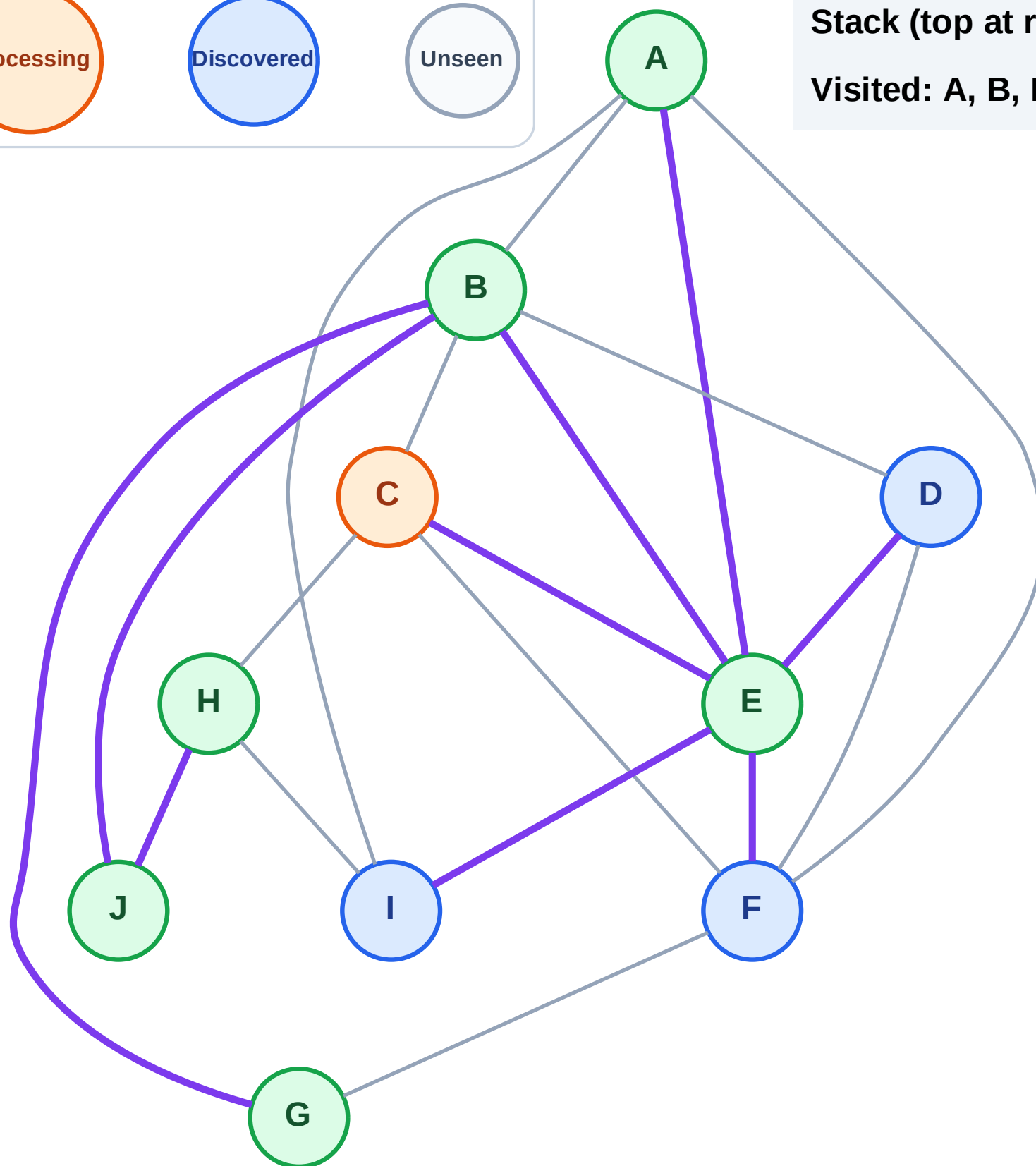
Step 14: Process node C

Legend



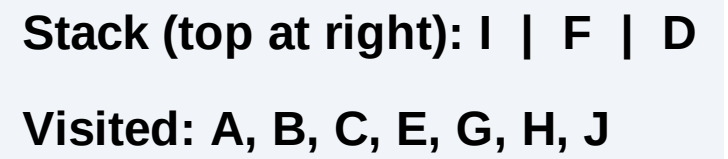
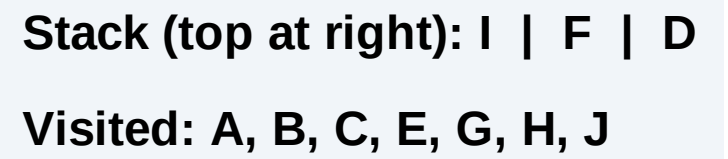
Stack (top at right): I | F | D

Visited: A, B, E, G, H, J



Step 15: No new neighbors from C

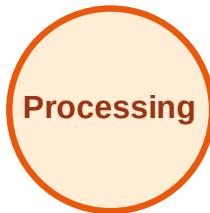
Legend



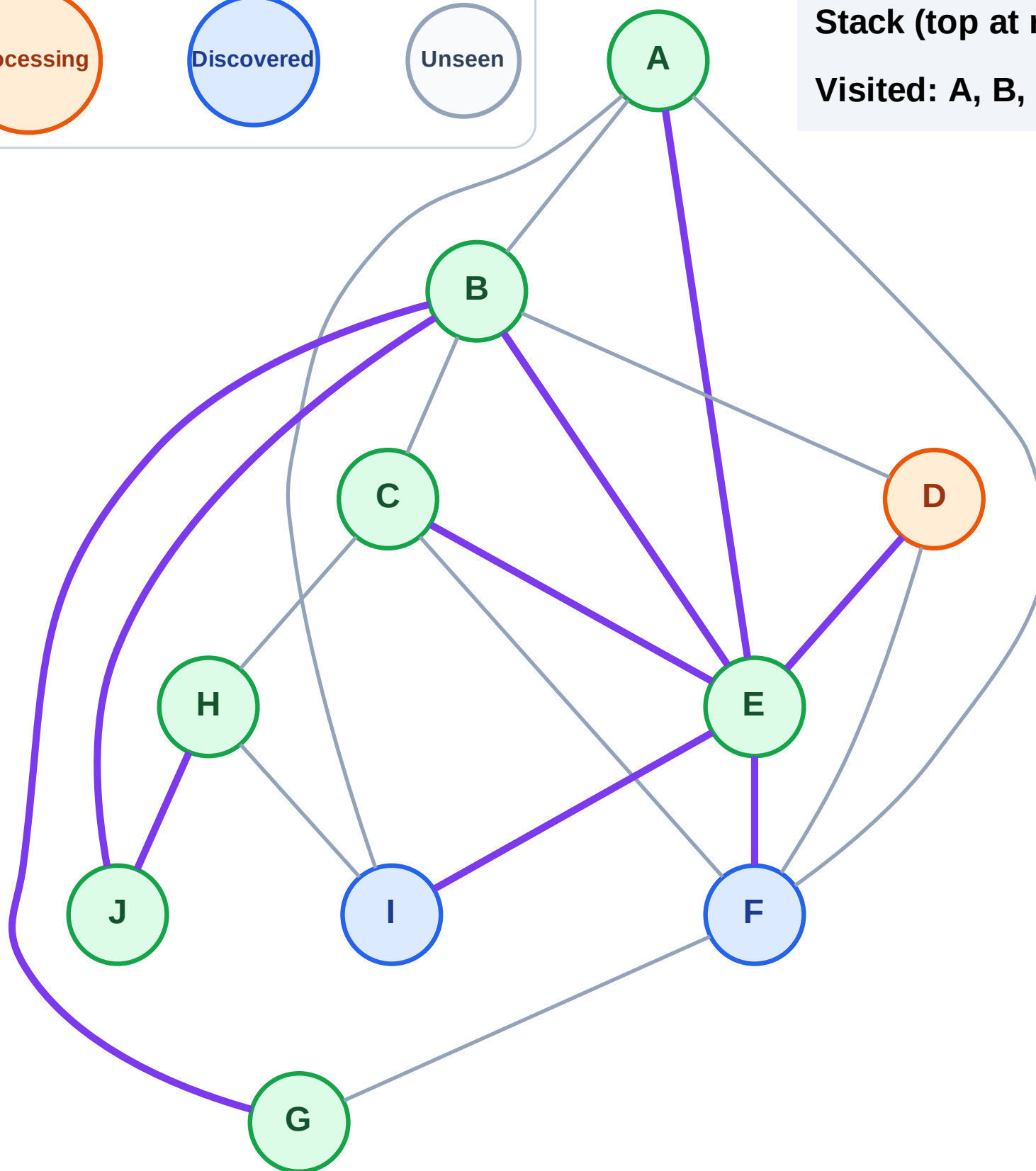
DFS Traversal

Step 16: Process node D

Legend



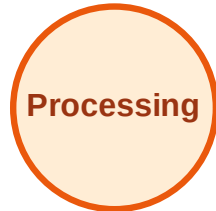
Stack (top at right): I | F
Visited: A, B, C, E, G, H, J



DFS Traversal

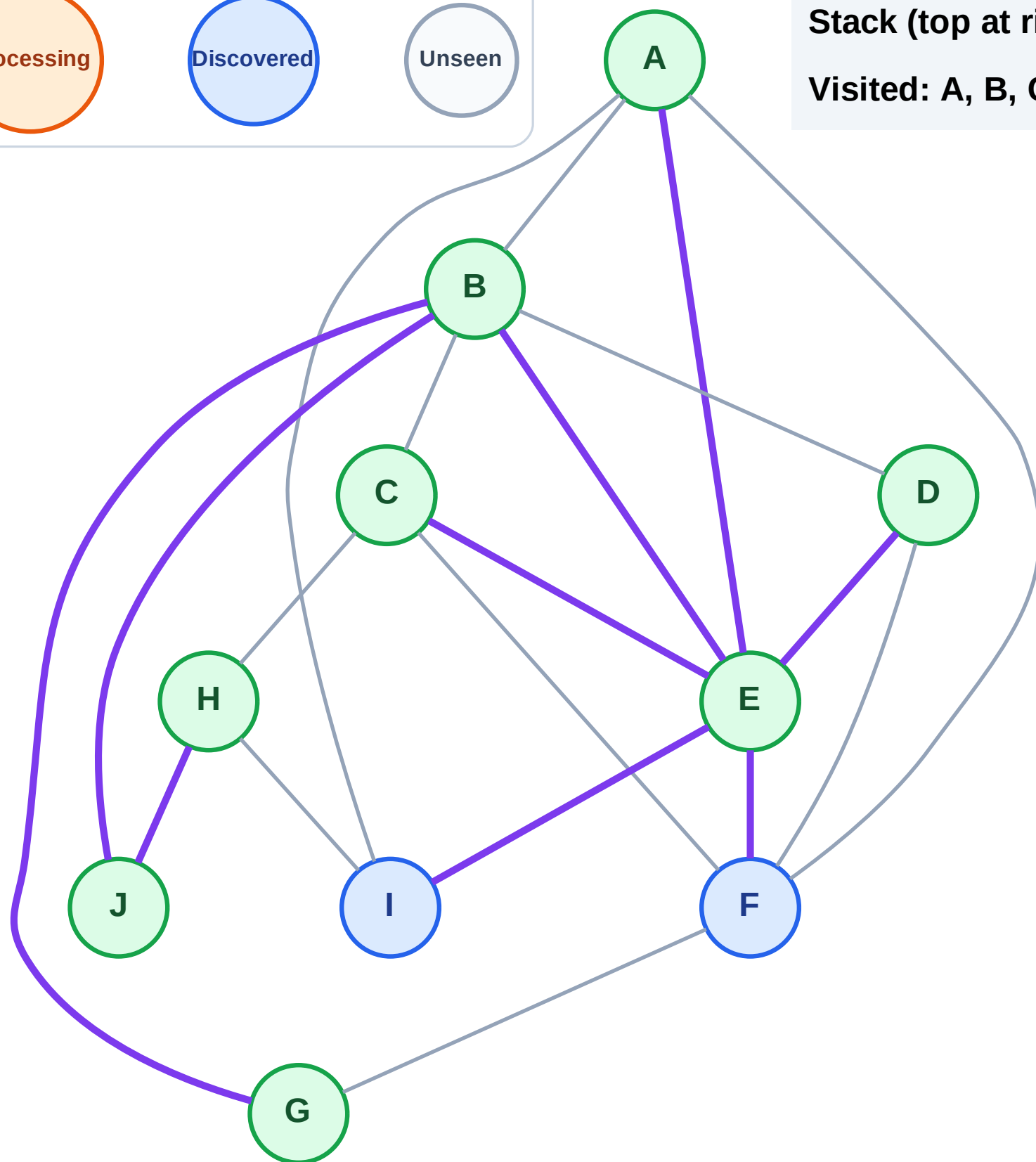
Step 17: No new neighbors from D

Legend



Stack (top at right): I | F

Visited: A, B, C, D, E, G, H, J



DFS Traversal

Step 18: Process node F

Legend

Complete

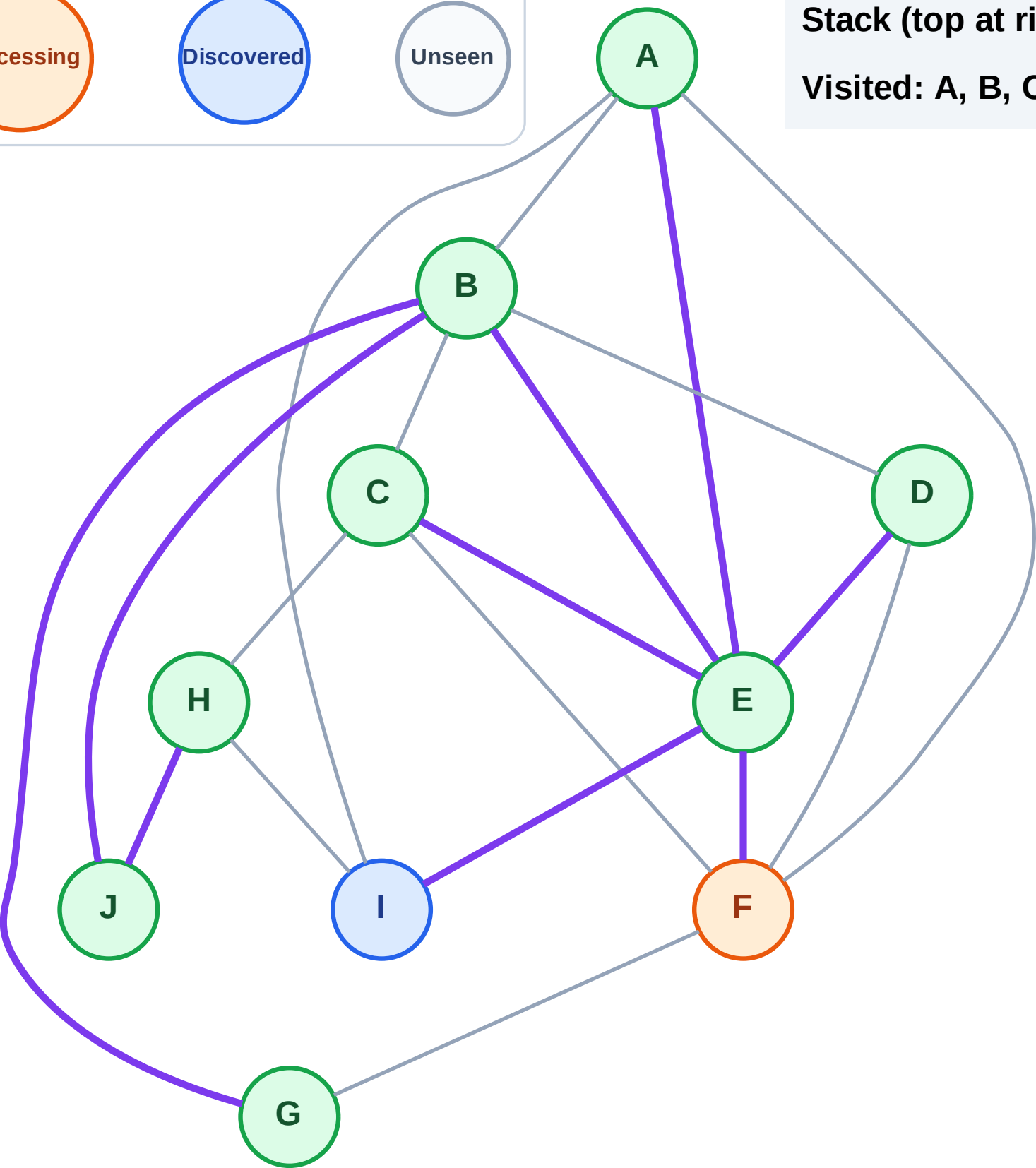
Processing

Discovered

Unseen

Stack (top at right): I

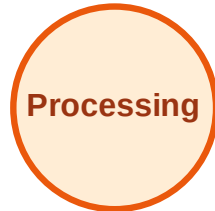
Visited: A, B, C, D, E, G, H, J



DFS Traversal

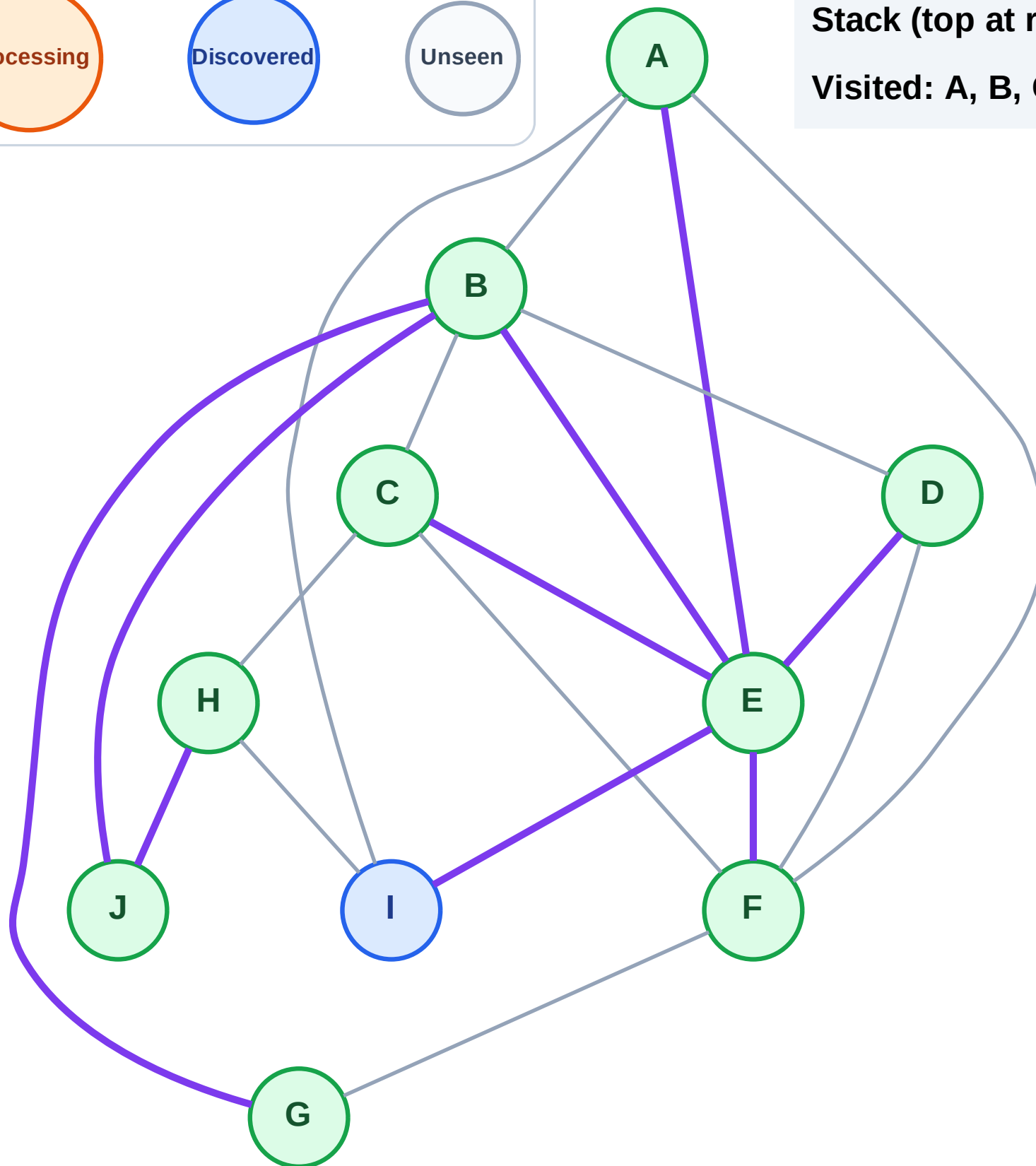
Step 19: No new neighbors from F

Legend



Stack (top at right): I

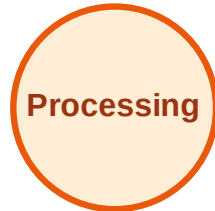
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

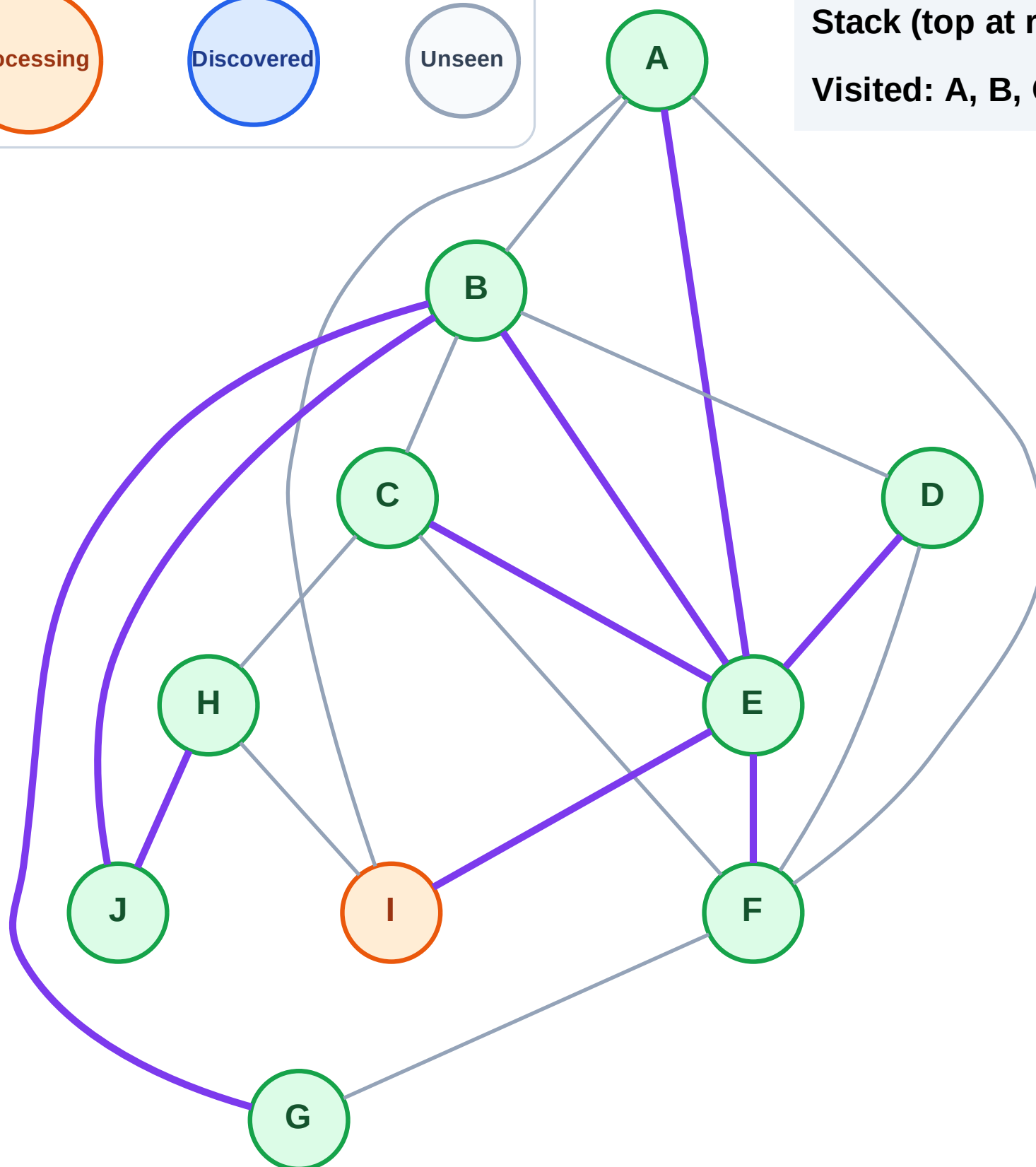
Step 20: Process node I

Legend



Stack (top at right): (empty)

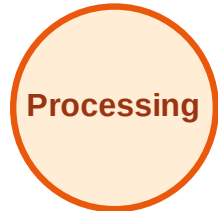
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

